UNIT 3 : THE DATA LINK LAYER

(8 HOURS)

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LECTURER

CONTENT

- Functions of Data Link Layer
- Data Link Control: Framing, Flow and Error Control
- Error detection and correction
- High-Level Data Link Control (HDLC) and point-to-point protocol (PPP)
- Channel allocation problem
- Multiple access: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Reservation, Polling, Token Passing), Channelization (FDMA, TDMA, CDMA)
- Wired LAN: Ethernet standards and FDDI
- Wireless LAN: IEEE 802.11 x and Bluetooth standards
- Token Bus, Token Ring and Virtual LAN

FUNCTION OF DATA LINK LAYER

- Data Link Layer is second layer of OSI Layered Model. This layer is one of the most complicated layers and has complex functionalities and liabilities.
- Data link layer hides the details of underlying hardware and represents itself to upper layer as the medium to communicate.
- Data link layer has two sub-layers:
  - **Logical Link Control**: It deals with protocols, flow-control, and error control
  - **Media Access Control**: It deals with actual control of media

FUNCTIONALITY OF DATA-LINK LAYER

- **Framing**
  - Data-link layer takes packets from Network Layer and encapsulates them into Frames.
  - Then, it sends each frame bit-by-bit on the hardware. At receivers end, data link layer picks up signals from hardware and assembles them into frames

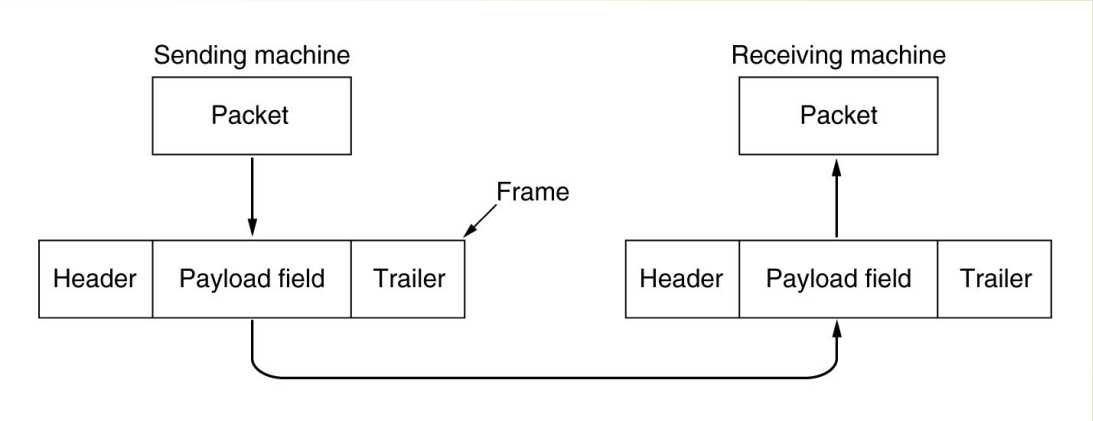
FUNCTION OF DATA LINK LAYER

- **Addressing**
  - Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.
- **Synchronization**
  - When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.
- **Error Control**
  - Sometimes signals may have encountered problem in transition and the bits are flipped. These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.
- **Flow Control**
  - Stations on same link may have different speed or capacity. Data-link layer ensures flow control that enables both machine to exchange data on same speed.
- **Multi-Access**
  - When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple Systems.

DATA LINK CONTROL: FRAMING

- The data link layer, needs to pack bits into frames, so that each frame is distinguishable from another.
- The Data Link layer prepares a packet for transport across the local media by encapsulating it with a header and a trailer to create a frame.
- The Data Link layer frame includes:
  - **Data** - The packet from the Network layer
  - **Header** - Contains control information, such as addressing, and is located at the beginning of the PDU (protocol data unit)
  - **Trailer** - Contains control information added to the end of the PDU

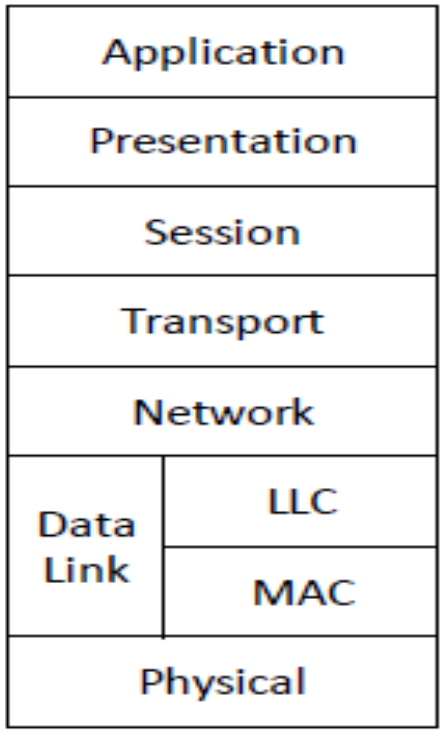
Parts of a Frame

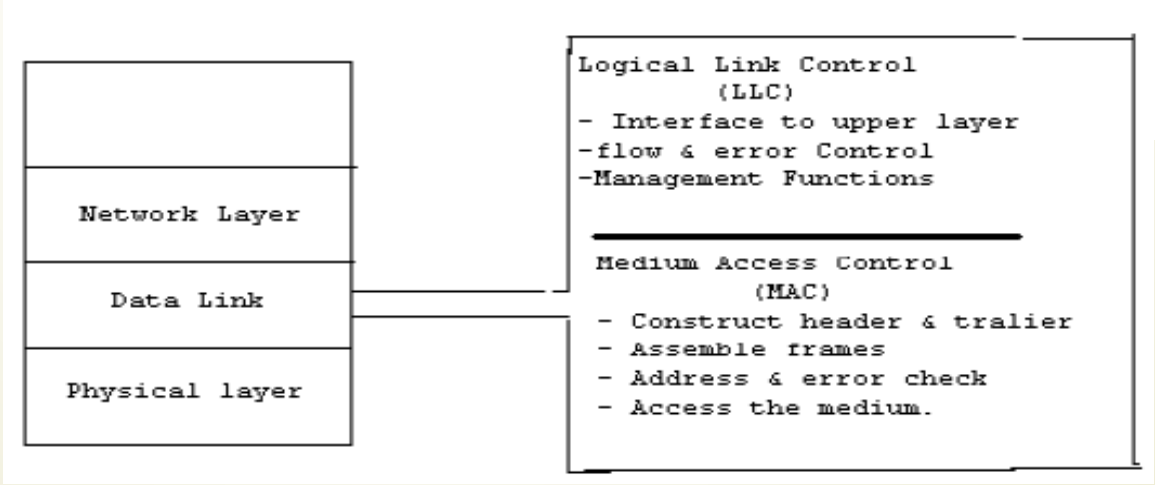


DATA LINK CONTROL: FRAMING

- **Frame Header** – It contains the source and the destination addresses of the frame.
- **Payload field** – It contains the message to be delivered.
- **Trailer** – It contains the error detection and error correction bits.
- **Flag** – It marks the beginning and end of the frame.

- Data link layer is divided into 2 sublayers
  - MAC (Media Access Control)
  - LLC (Logical Link Control)





DATA LINK CONTROL: FRAMING

DSAP	SSAP	Control	Data

Fig :- LLC Frame Format

- **Destination Service Access Point (DSAP)**
  - IEEE 802.2 header begins with a 1 byte field, which identifies the receiving upper-layer process.
- **Source Service Access Point (SSAP)**
  - Following the DSAP address is the 1-byte address, which identifies the sending upper-layer process.
- **Control** –
  - The Control field employs three different formats, depending on the type of LLC frame used:
  - **Information (I) frame**
    - Carries upper-layer information and some control information.

DATA LINK CONTROL: FRAMING

- **Supervisory (S) frame**
  - Provides control information. An S frame can request and suspend transmission, reports on status, and acknowledge receipt of I frames. S frames do not have an Information field.
- **Unnumbered (U) frame**
  - Used for control purposes and is not sequenced. A U frame can be used to initialize secondaries. Depending on the function of the U frame, its Control field is 1 or 2 bytes. Some U frames have an Information field.

- **Data**
  - Variable-length field bounded by the MAC format implemented. Usually contains IEEE 802.2 Subnetwork Access Protocol (SNAP) header information, as well as application-specific data.

Control Header	Source Addrerss	Destination Address	LLC Data	CRC

Fig : General MAC frame Format

DATA LINK CONTROL: FRAMING

Ethernet Protocol

A Common Data Link Layer Protocol for LANs

Frame

Field name	Preamble	Destination	Source	Type	Data	Frame Check Sequence
Size	8 bytes	6 bytes	6 bytes	2 bytes	46 - 1500 bytes	4 bytes

Preamble - used for synchronization; also contains a delimiter to mark the end of the timing information.

Destination Address - 48 bit MAC address for the destination node.

Source Address - 48 bit MAC address for the source node.

Type - value to indicate which upper layer protocol will receive the data after the Ethernet process is complete.

Data or payload - this is the PDU, typically an IPv4 packet, that is to be transported over the media.

Frame Check Sequence (FCS) - A value used to check for damaged frames.

DATA LINK CONTROL: FRAMING

Problems in Framing –

- Detecting start of the frame: SFD (Starting Frame Delimiter).
- How do station detect a frame: Station checks destination address to accept or reject frame.
- Detecting end of frame: When to stop reading the frame.

Types of framing

- There are two types of framing:
  - **Fixed-sized Framing**
    - Here the size of the frame is fixed so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame.
    - Example – ATM cells.

DATA LINK CONTROL: FRAMING

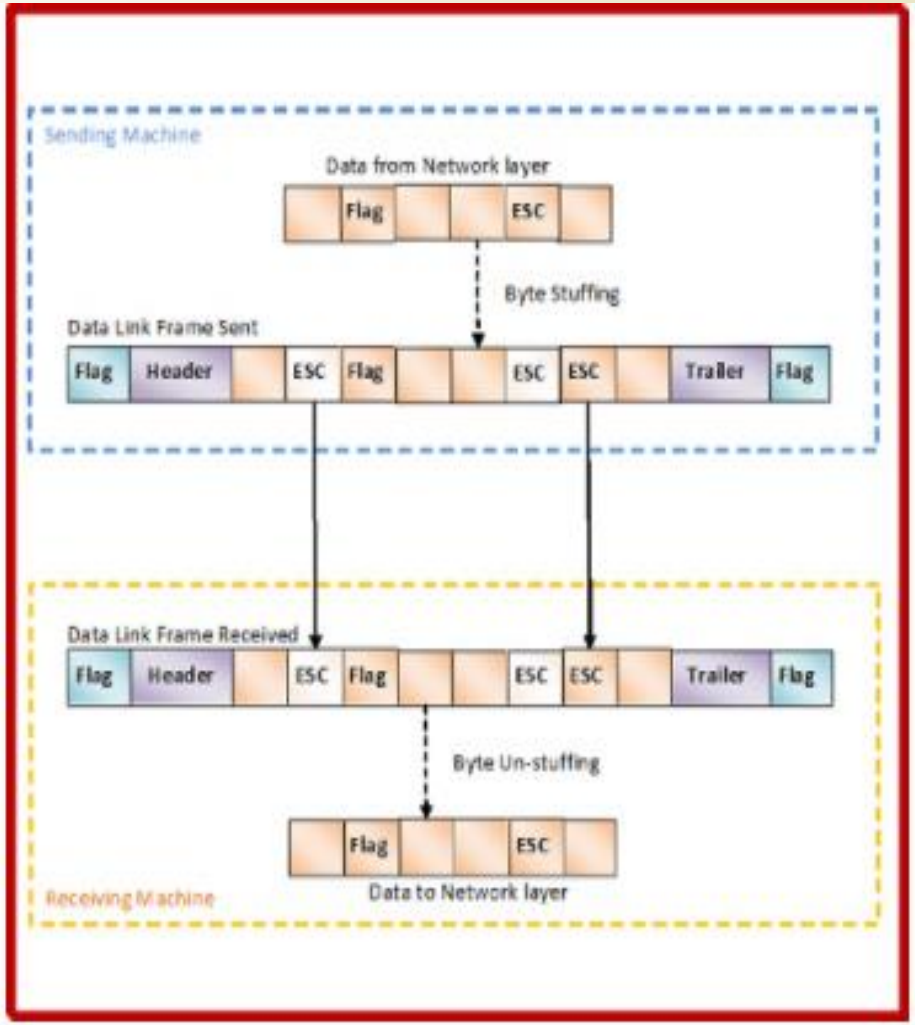
- **Variable – Sized Framing**
  - Here, the size of each frame to be transmitted may be different. So additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.
  - It is used in local area networks.
  - Two ways to define frame delimiters in variable sized framing are −
    - **Length Field** − Here, a length field is used that determines the size of the frame. It is used in Ethernet (IEEE 802.3).
    - **End Delimiter** − Here, a pattern is used as a delimiter to determine the size of frame. It is used in Token Rings. If the pattern occurs in the message, then two approaches are used to avoid the situation −
      - **Byte – Stuffing** − A byte is stuffed in the message to differentiate from the delimiter. This is also called character-oriented framing.
      - **Bit – Stuffing** − A pattern of bits of arbitrary length is stuffed in the message to differentiate from the delimiter. This is also called bit – oriented framing.

DATA LINK CONTROL: FRAMING

1. Byte stuffing

- If the pattern of the flag byte is present in the message byte, there should be a strategy so that the receiver does not consider the pattern as the end of the frame.
- In character – oriented protocol, the mechanism adopted is byte stuffing.

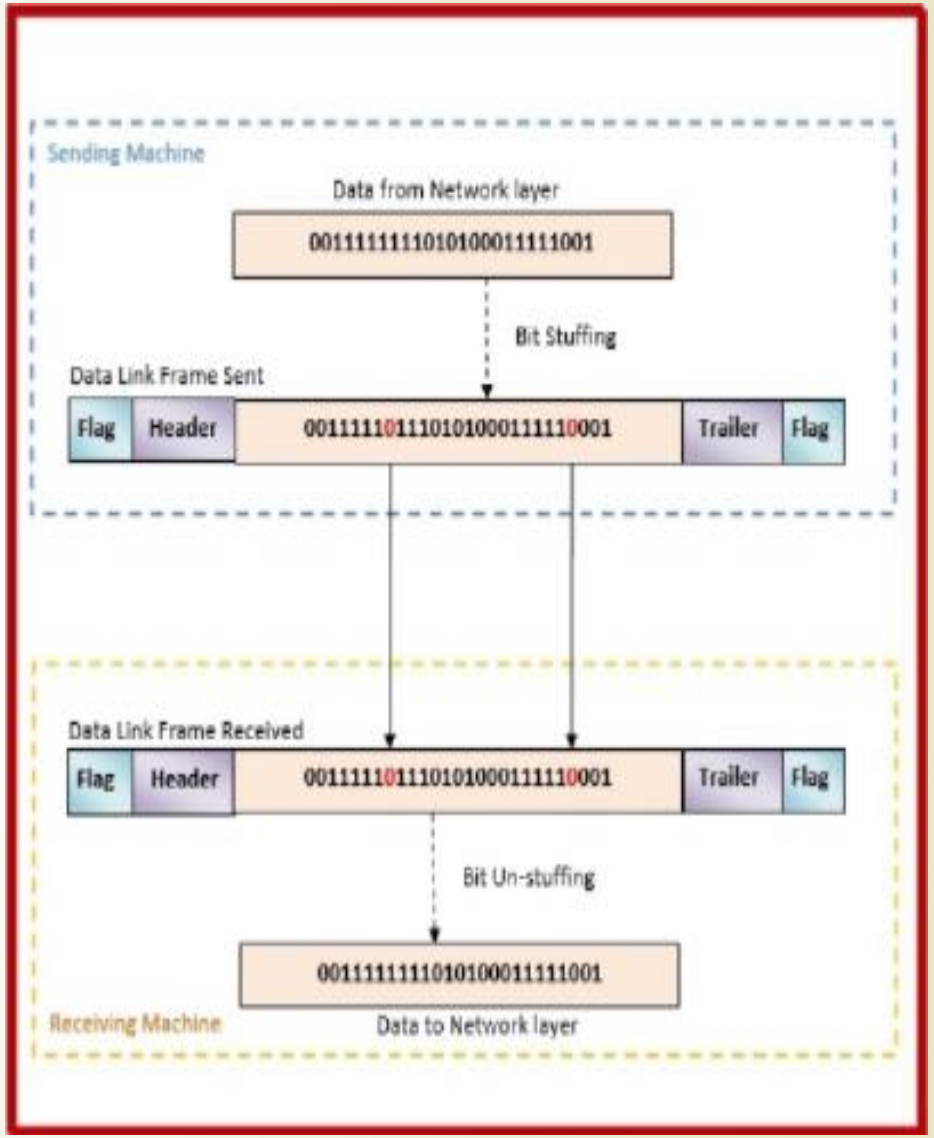
- In byte stuffing, a special byte called the escape character (ESC) is stuffed before every byte in the message with the same pattern as the flag byte.
- If the ESC sequence is found in the message byte, then another ESC byte is stuffed before it.



DATA LINK CONTROL: FRAMING

2. Bit stuffing

- In a data link frame, the delimiting flag sequence generally contains six or more consecutive 1s.
- In order to differentiate the message from the flag in case of the same sequence, a single bit is stuffed in the message.
- Whenever a 0 bit is followed by five consecutive 1bits in the message, an extra 0 bit is stuffed at the end of the five 1s.
- When the receiver receives the message, it removes the stuffed 0s after each sequence of five 1s. The un-stuffed message is then sent to the upper layers.



DATA LINK CONTROL: FLOW CONTROL

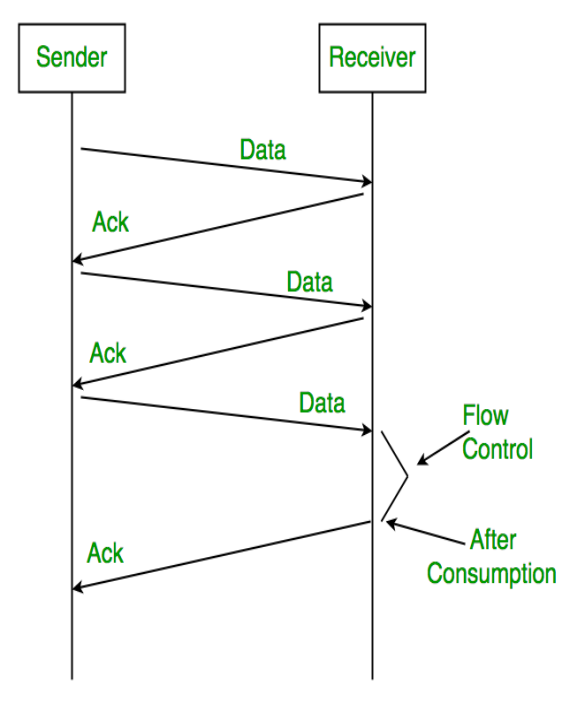
- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It is a set of measures taken to regulate the amount of data that a sender sends so that a fast sender does not overwhelm a slow receiver.
- In data link layer, flow control restricts the number of frames the sender can send before it waits for an acknowledgment from the receiver.

Approaches of Flow Control

- Flow control can be broadly classified into two categories −
  - **Feedback based Flow Control** In these protocols, the sender sends frames after it has received acknowledgments from the user. This is used in the data link layer.
  - **Rate based Flow Control** These protocols have built in mechanisms to restrict the rate of transmission of data without requiring acknowledgment from the receiver. This is used in the network layer and the transport layer.

DATA LINK CONTROL: FLOW CONTROL

Flow Control Techniques in Data Link Layer



Stop and Wait

- This protocol involves the following transitions −
  - The sender sends a frame and waits for acknowledgment.
  - Once the receiver receives the frame, it sends an acknowledgment frame back to the sender.
  - On receiving the acknowledgment frame, the sender understands that the receiver is ready to accept the next frame. So it sender the next frame in queue.

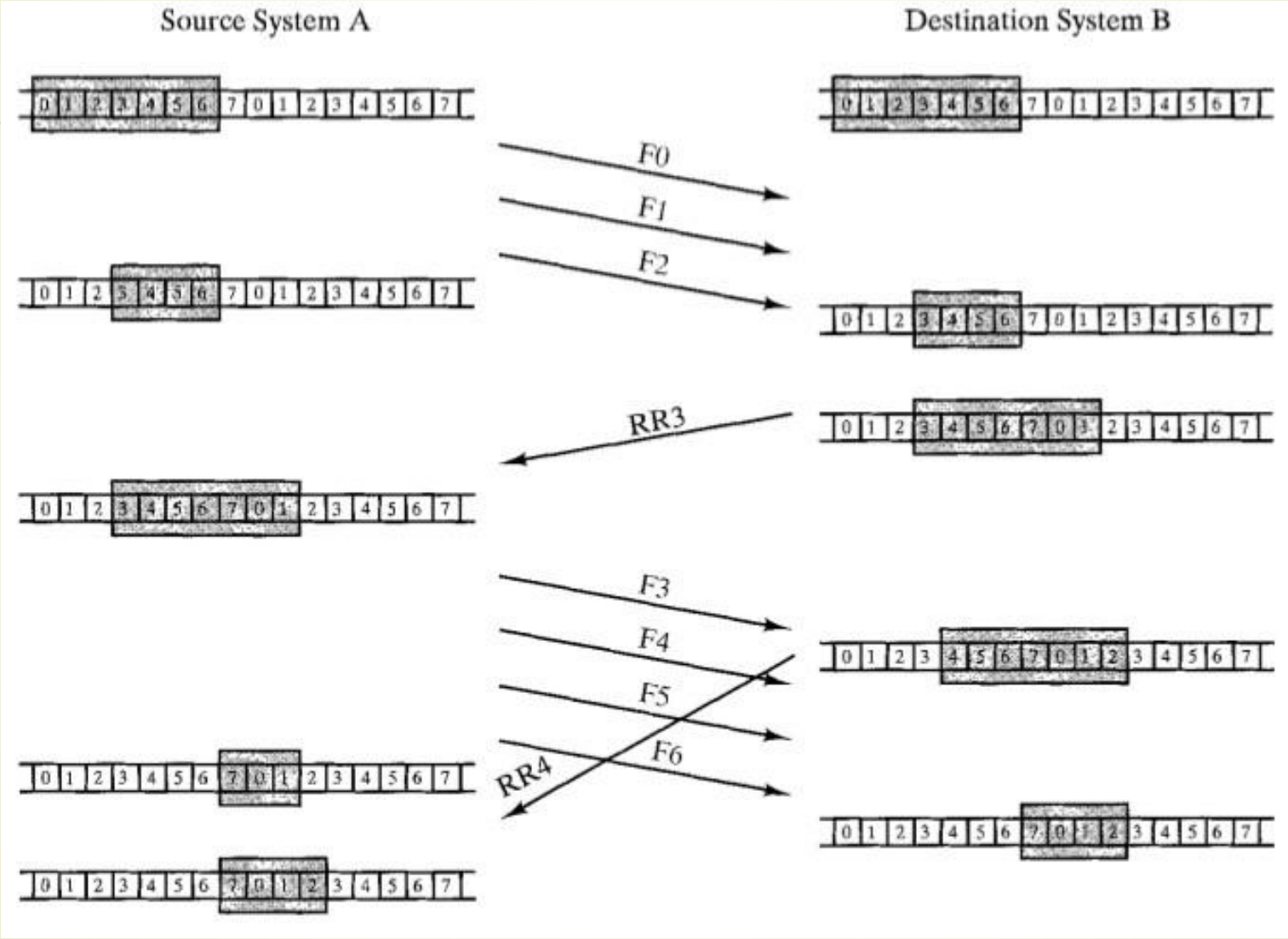
DATA LINK CONTROL: FLOW CONTROL

Sliding Window

- This protocol improves the efficiency of stop and wait protocol by allowing multiple frames to be transmitted before receiving an acknowledgment.
- The working principle of this protocol can be described as follows −
  - Both the sender and the receiver has finite sized buffers called windows.
  - The sender and the receiver agrees upon the number of frames to be sent based upon the buffer size.
  - The sender sends multiple frames in a sequence, without waiting for acknowledgment.
  - When its sending window is filled, it waits for acknowledgment.
  - On receiving acknowledgment, it advances the window and transmits the next frames, according to the number of acknowledgments received.
  - The example assumes a 3-bit sequence number field and a maximum window size of seven frames.

DATA LINK CONTROL: FLOW CONTROL

- The example assumes a 3-bit sequence number field and a maximum window size of seven frames.



DATA LINK CONTROL: ERROR CONTROL

- Error control in data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission.
- In case of lost or corrupted frames, the receiver does not receive the correct data-frame and sender is ignorant about the loss. Data link layer follows a technique to detect transit errors and take necessary actions, which is retransmission of frames whenever error is detected or frame is lost. The process is called Automatic Repeat Request (ARQ)

Phases in Error Control

- The error control mechanism in data link layer involves the following phases −
  - **Detection of Error** − Transmission error, if any, is detected by either the sender or the receiver.
  - **Acknowledgment** − acknowledgment may be positive or negative.
    - **Positive ACK** − On receiving a correct frame, the receiver sends a positive acknowledge.
    - **Negative ACK** − On receiving a damaged frame or a duplicate frame, the receiver sends a negative acknowledgment back to the sender.

DATA LINK CONTROL: ERROR CONTROL

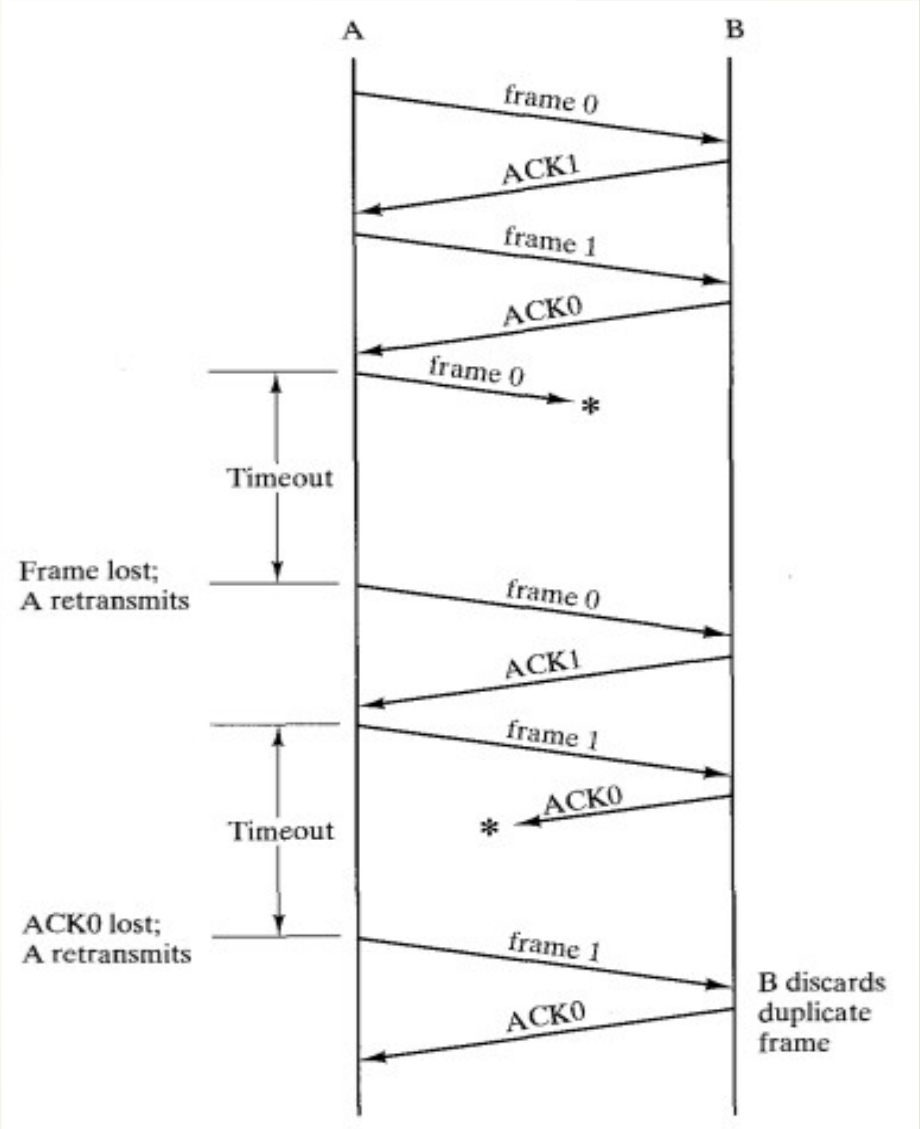
- **Retransmission** − The sender maintains a clock and sets a timeout period. If an acknowledgment of a data-frame previously transmitted does not arrive before the timeout, or a negative acknowledgment is received, the sender retransmits the frame.

Error Control Techniques

DATA LINK CONTROL: ERROR CONTROL

Stop and Wait ARQ

- This protocol involves the following transitions −
  - A timeout counter is maintained by the sender, which is started when a frame is sent.
  - If the sender receives acknowledgment of the sent frame within time, the sender is confirmed about successful delivery of the frame. It then transmits the next frame in queue.
  - If the sender does not receive the acknowledgment within time, the sender assumes that either the frame or its acknowledgment is lost in transit. It then retransmits the frame.
  - If the sender receives a negative acknowledgment, the sender retransmits the frame.



DATA LINK CONTROL: ERROR CONTROL

Bidirectional Transmission:

- The stop-and-wait mechanism we have discussed is unidirectional.
- However, we can have bi-directional transmission if the two parties have two separate channels for the full-duplex transmission or share the same channel for half-duplex transmission.

Piggybacking:

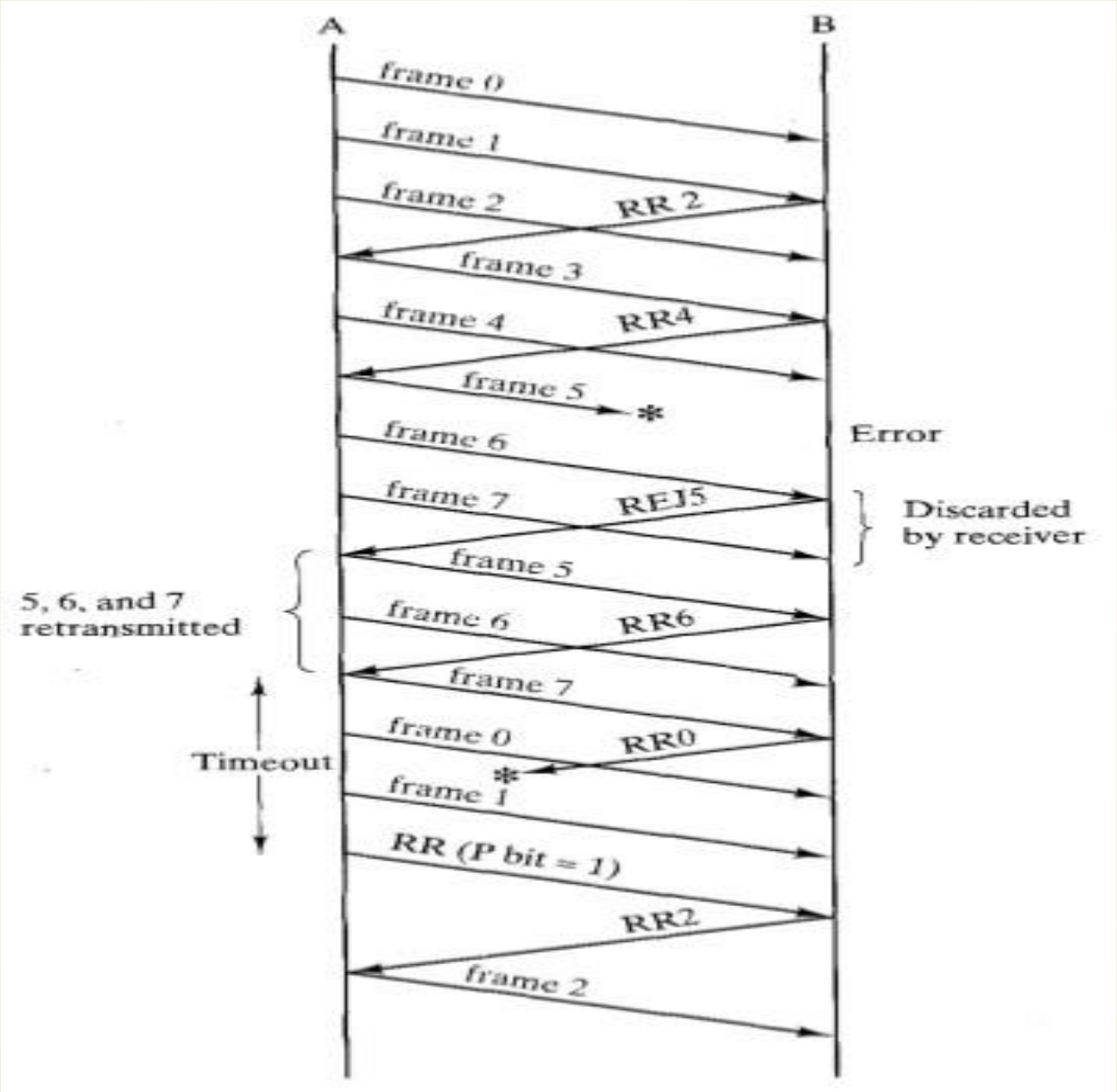
- It is a method to combine a data frame with an acknowledgement.
- For example, stations A and B both have data to send. Instead of sending separate data and ACK frames, Station A sends a data frame that includes an ACK , station B behaves in a similar manner.
- Piggybacking can save BW because the overhead from a data frame and ACK frame (addresses, CRC, etc) can be combined into just one frame.

DATA LINK CONTROL: ERROR CONTROL

Go-Back-N ARQ

- The working principle of this protocol is −
  - The sender has buffers called sending window.
  - The sender sends multiple frames based upon the sending-window size, without receiving the acknowledgment of the previous ones.
  - The receiver receives frames one by one. It keeps track of incoming frame's sequence number and sends the corresponding acknowledgment frames.
  - After the sender has sent all the frames in window, it checks up to what sequence number it has received positive acknowledgment (RR- receive request).
  - If the sender has received positive acknowledgment for all the frames, it sends next set of frames.
  - If sender receives REJ (reject) (NACK) or has not receive any ACK for a particular frame, it retransmits all the frames after which it does not receive any positive ACK.

DATA LINK CONTROL: ERROR CONTROL

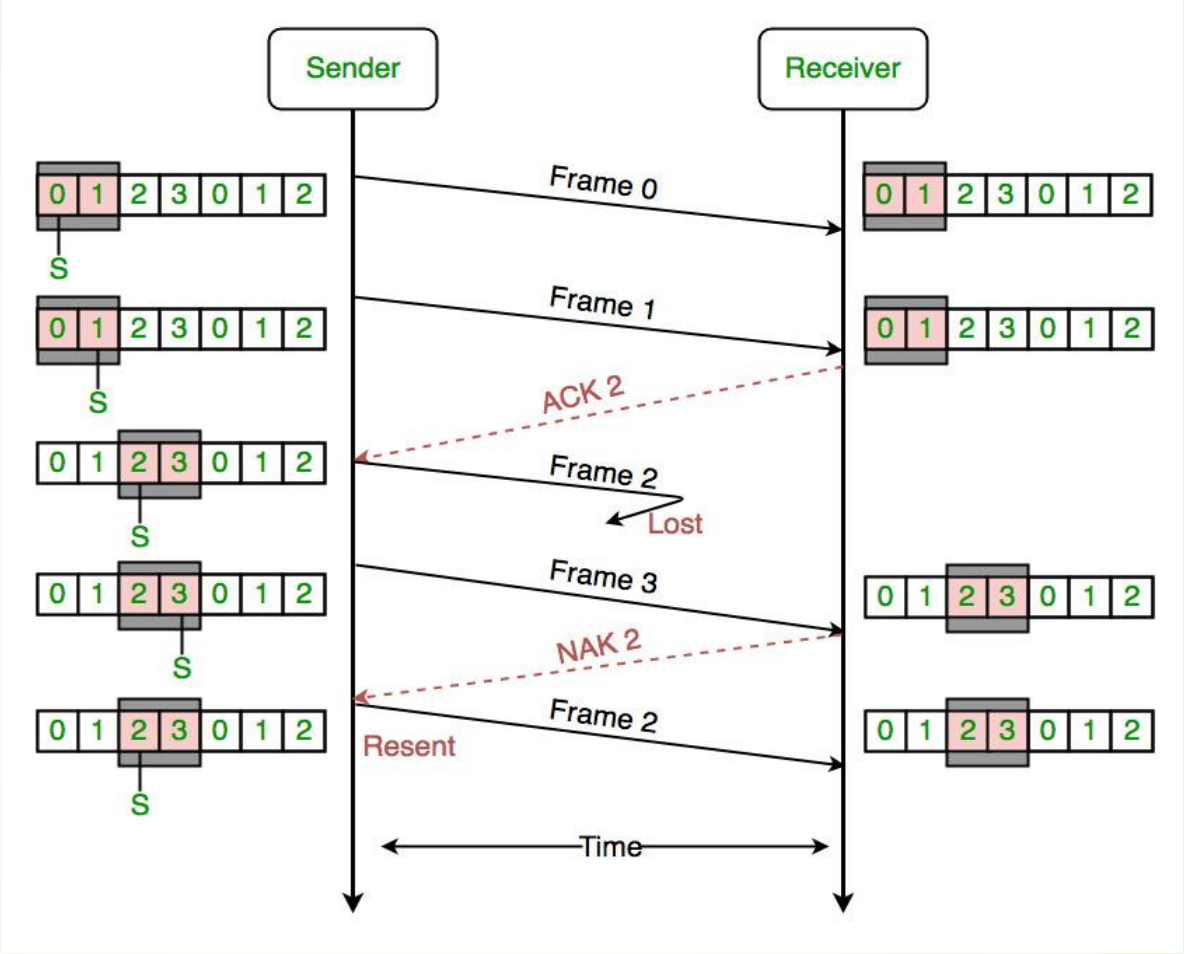


DATA LINK CONTROL: ERROR CONTROL

Selective Repeat ARQ

- Both the sender and the receiver have buffers called sending window and receiving window respectively.
- The sender sends multiple frames based upon the sending-window size, without receiving the acknowledgment of the previous ones.
- The receiver also receives multiple frames within the receiving window size.
- The receiver keeps track of incoming frame's sequence numbers, buffers the frames in memory.
- It sends ACK for all successfully received frames and sends NACK for only frames which are missing or damaged.
- The sender in this case, sends only packet for which NACK is received.

DATA LINK CONTROL: ERROR CONTROL



ERROR DETECTION AND CORRECTION

Error

- A condition when the receiver's information does not match with the sender's information.
- During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0.

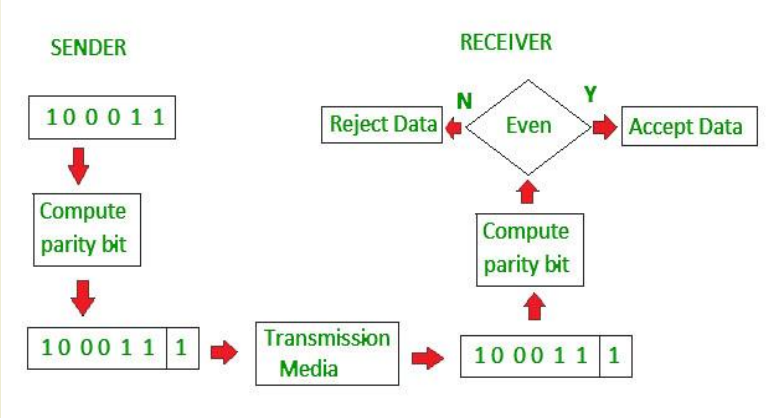
Error Detecting Codes (Implemented either at Data link layer or Transport Layer of OSI Model)

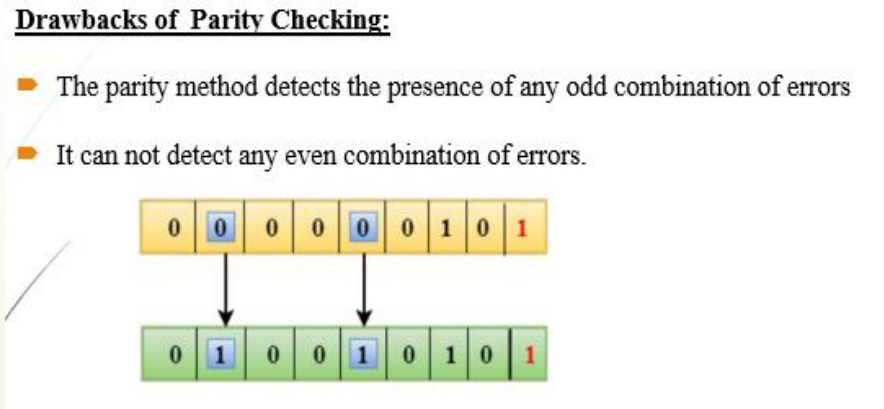
- Whenever a message is transmitted, it may get scrambled by noise or data may get corrupted.
- To avoid this, we use error-detecting codes which are additional data added to a given digital message to help us detect if any error has occurred during transmission of the message.
- Basic approach used for error detection is the use of redundancy bits, where additional bits are added to facilitate detection of errors.
- Some popular techniques for error detection are:
  - Simple Parity check

ERROR DETECTION AND CORRECTION

- Two-dimensional Parity check
- Checksum
- Cyclic redundancy check

- **Simple Parity check**
  - Blocks of data from the source are subjected to a check bit or parity bit generator form, where a parity of :
    - 1 is added to the block if it contains odd number of 1's, and
    - 0 is added if it contains even number of 1's
  - This scheme makes the total number of 1's even, that is why it is called even parity checking





ERROR DETECTION AND CORRECTION

Two-dimensional Parity check

- Parity check bits are calculated for each row, which is equivalent to a simple parity check bit.
- Parity check bits are also calculated for all columns, then both are sent along with the data.
- At the receiving end these are compared with the parity bits calculated on the received data.

Original Data

10011001	11100010	00100100	10000100

	Row parities
10011001	0
11100010	0
00100100	0
10000100	0
11011011	0

Column parities

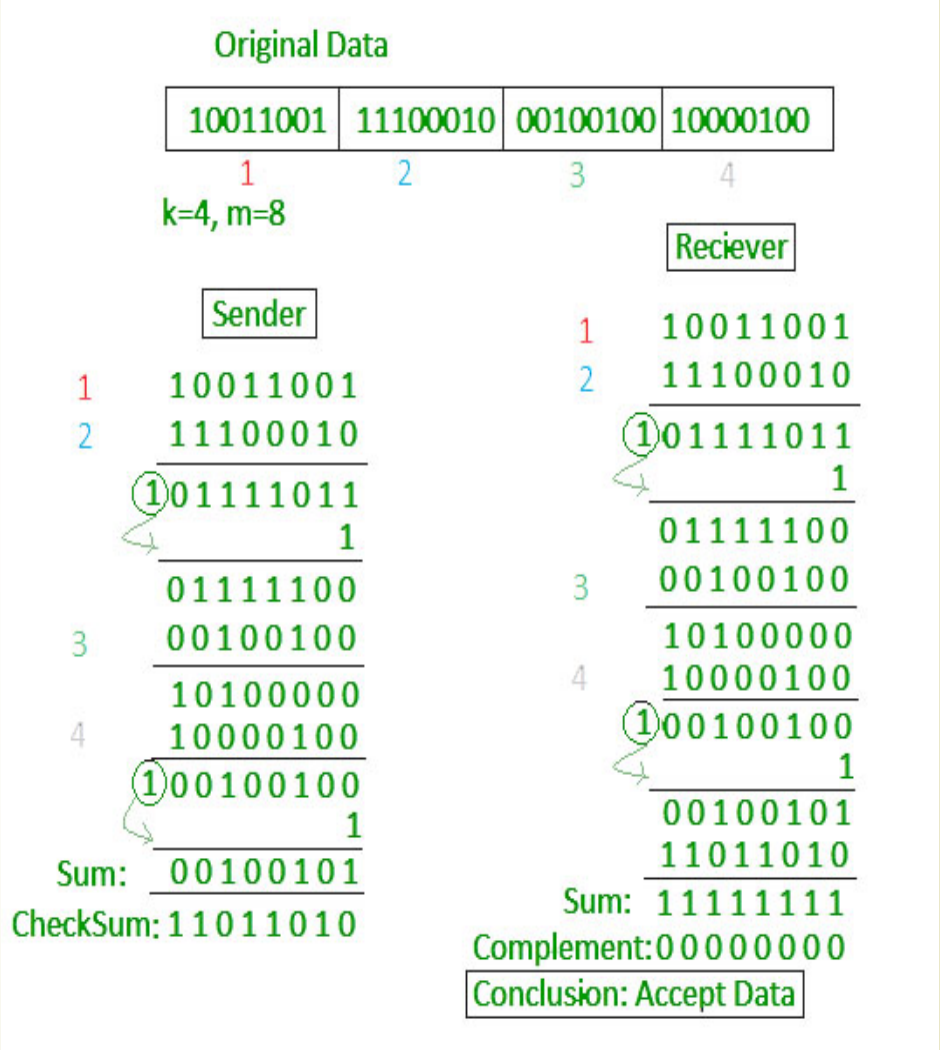
100110010	111000100	001001000	100001000	110110110

Data to be sent

ERROR DETECTION AND CORRECTION

Checksum

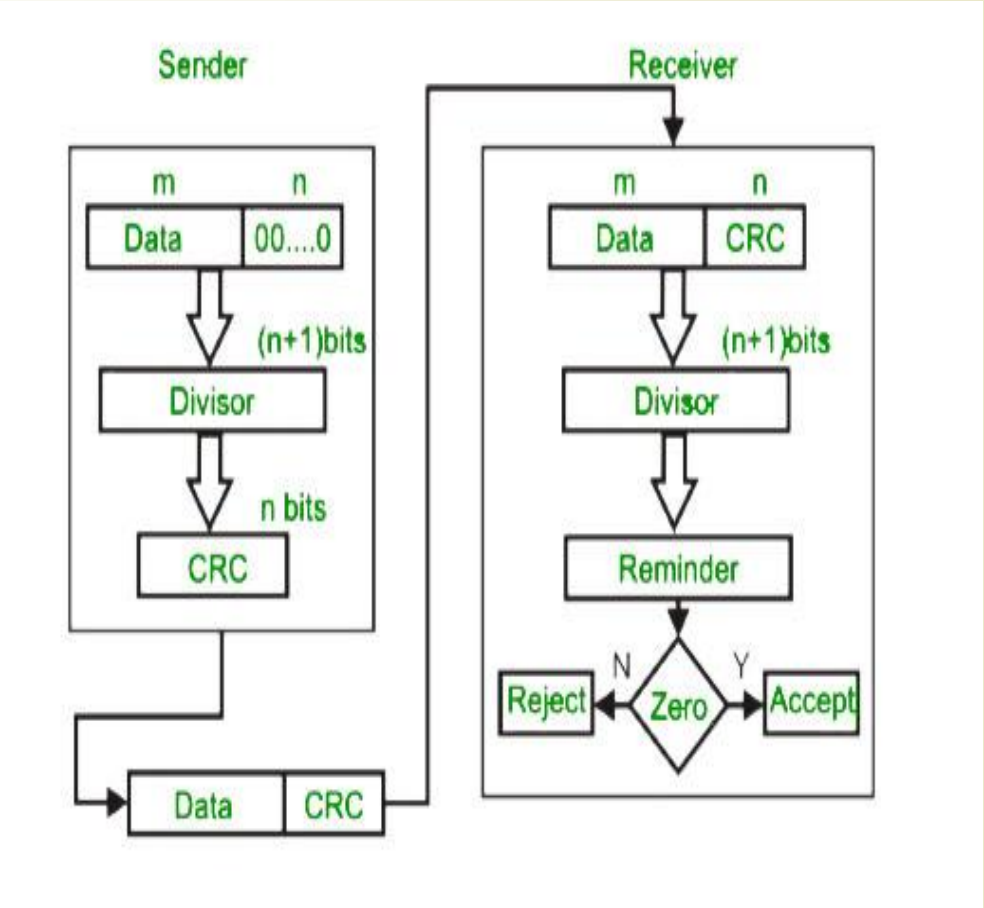
- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



ERROR DETECTION AND CORRECTION

Cyclic redundancy check (CRC)

- Unlike checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.



ERROR DETECTION AND CORRECTION

original message
1 0 1 0 0 0 0

@ means X-OR

Generator polynomial
x^3+1
$1.x^3+0.x^2+0.x^1+1.x^0$
CRC generator
1 0 0 1 4-bit

If CRC generator is of n bit then append (n-1) zeros in the end of original message

ERROR DETECTION AND CORRECTION

Data word to be sent - 100100
Key - 1101 [Or generator polynomial $x^3 + x^2 + 1$]
Sender Side:

```
          111101
1101 | 100100000 |
       1101
       ----
        1000
        1101
        ----
         1010
         1101
         ----
          1110
          1101
          ----
           0110
           0000
           ----
            1100
            1101
            ----
             001
```

Therefore, the remainder is 001 and hence the encoded data sent is 100100001.

Receiver Side:
Code word received at the receiver side 100100001

```
          111101
1101 | 100100001 |
       1101
       ----
        1000
        1101
        ----
         1010
         1101
         ----
          1110
          1101
          ----
           0110
           0000
           ----
            1101
            1101
            ----
             0000
```

Therefore, the remainder is all zeros. Hence, the data received has no error

ERROR DETECTION AND CORRECTION

33 Q. A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1. What is the actual bit string transmitted?

Solution-

- The generator polynomial $G(x) = x^4 + x + 1$ is encoded as 10011. (i.e. $x^4x^3\ x^2\ x^1x^0$)
- Clearly, the generator polynomial consists of 5 bits.
- So, a string of 4 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 11010110110000.

Now, the binary division is performed as-

```
                1 1 0 0 0 0 1 0 1 0
10011 | 1 1 0 1 0 1 1 0 1 1 0 0 0 0
        1 0 0 1 1
          1 0 0 1 1
          1 0 0 1 1
            0 0 0 0 1
            0 0 0 0 0
              0 0 0 1 0
              0 0 0 0 0
                0 0 1 0 1
                0 0 0 0 0
                  0 1 0 1 1
                  0 0 0 0 0
                    1 0 1 1 0
                    1 0 0 1 1
                      0 1 0 1 0
                      0 0 0 0 0
                        1 0 1 0 0
                        1 0 0 1 1
                          0 1 1 1 0
                          0 0 0 0 0
                            1 1 1 0   <- Remainder
```

From here, CRC = 1110.

Now,

- The code word to be transmitted is obtained by replacing the last 4 zeroes of 11010110110000 with the CRC.
- Thus, the code word transmitted to the receiver = 11010110111110.

ERROR DETECTION AND CORRECTION

ERROR CORRECTING CODE:

- Error Correction codes are used to detect and correct the errors when data is transmitted from the sender to the receiver.
- Error Correction can be handled in two ways:
  - **Backward error correction:** Once the error is discovered, the receiver requests the sender to retransmit the entire data unit.
  - **Forward error correction**: In this case, the receiver uses the error-correcting code which automatically corrects the errors.
- A single additional bit can detect the error, but cannot correct it.
- Suppose r is the number of redundant bits and d is the total number of the data bits. The number of redundant bits r can be calculated by using the formula:
- $2^r >= d+r+1$

ERROR DETECTION AND CORRECTION

Hamming Code:

Algorithm of Hamming code:

- An information of 'd' bits are added to the redundant bits 'r' to form d+r.
- The location of each of the (d+r) digits is assigned a decimal value.
- The 'r' bits are placed in the positions $1,2,.....2^{k-1}$.
- At the receiving end, the parity bits are recalculated. The decimal value of the parity bits determines the position of an error.

Suppose a binary data 1001101 is to be transmitted. To implement hamming code for this, following steps are used:

1. Calculating the number of redundancy bits required. Since number of data bits is 7, the value of r is calculated as

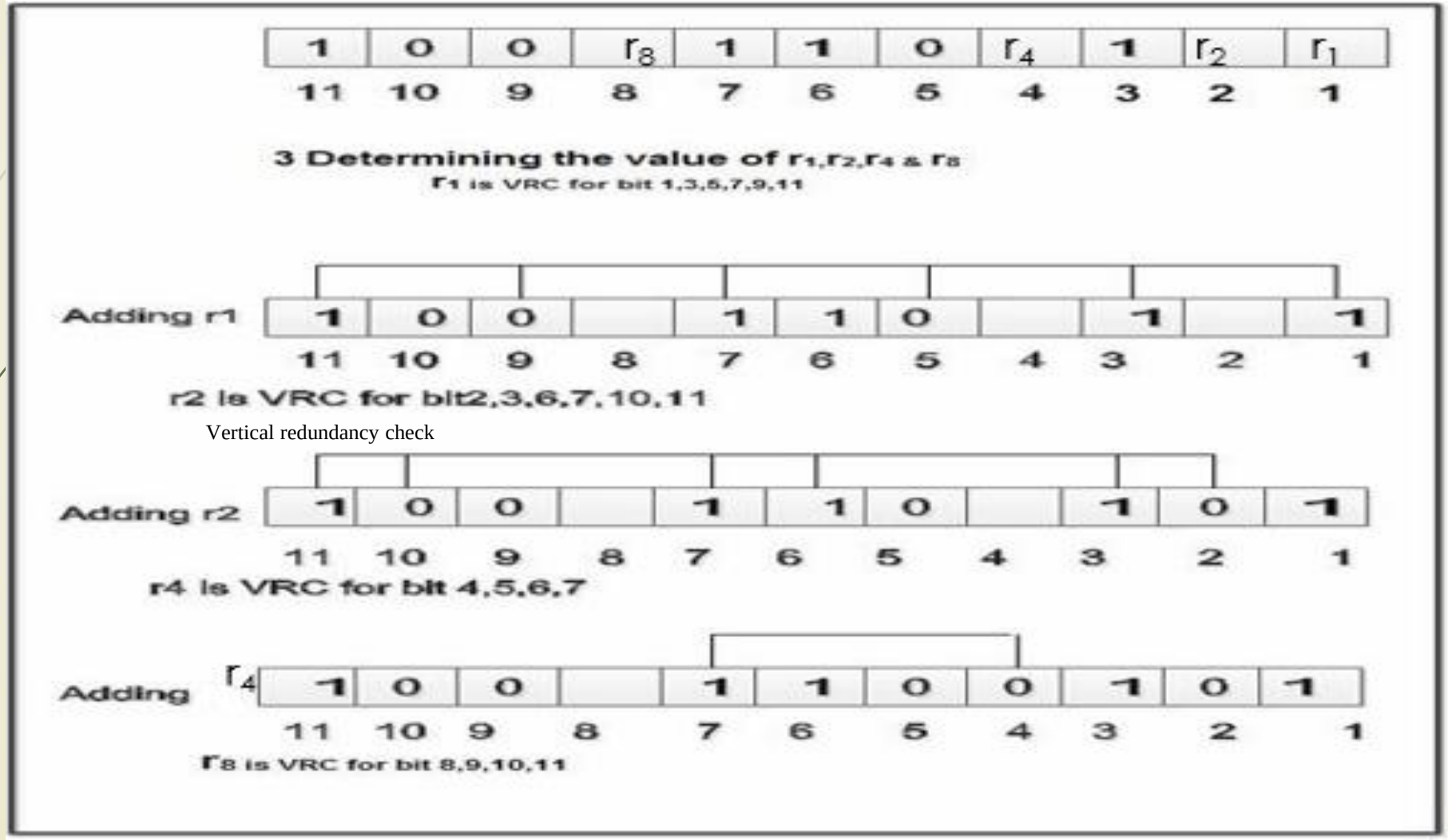
$2^r > m + r + 1$

$2^4 > 7 + 4 + 1$

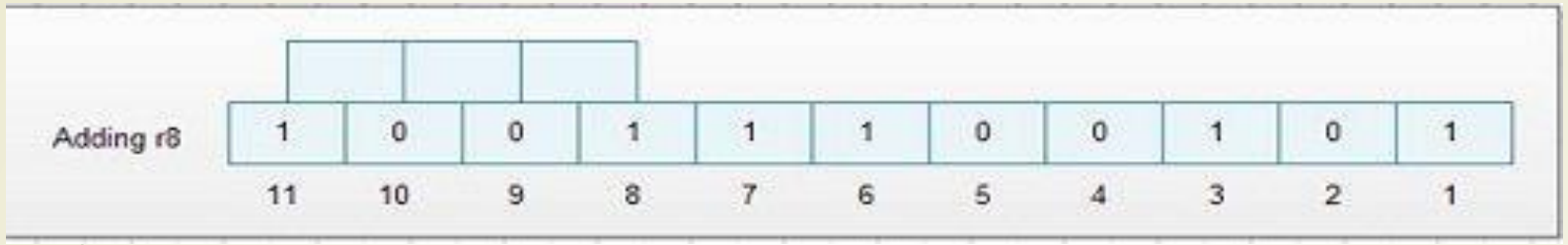
Therefore no. of redundancy bits = 4

ERROR DETECTION AND CORRECTION

2. Determining the positions of various data bits and redundancy bits. The various r bits are placed at the position that corresponds to the power of 2 *i.e.* 1, 2, 4, 8



ERROR DETECTION AND CORRECTION

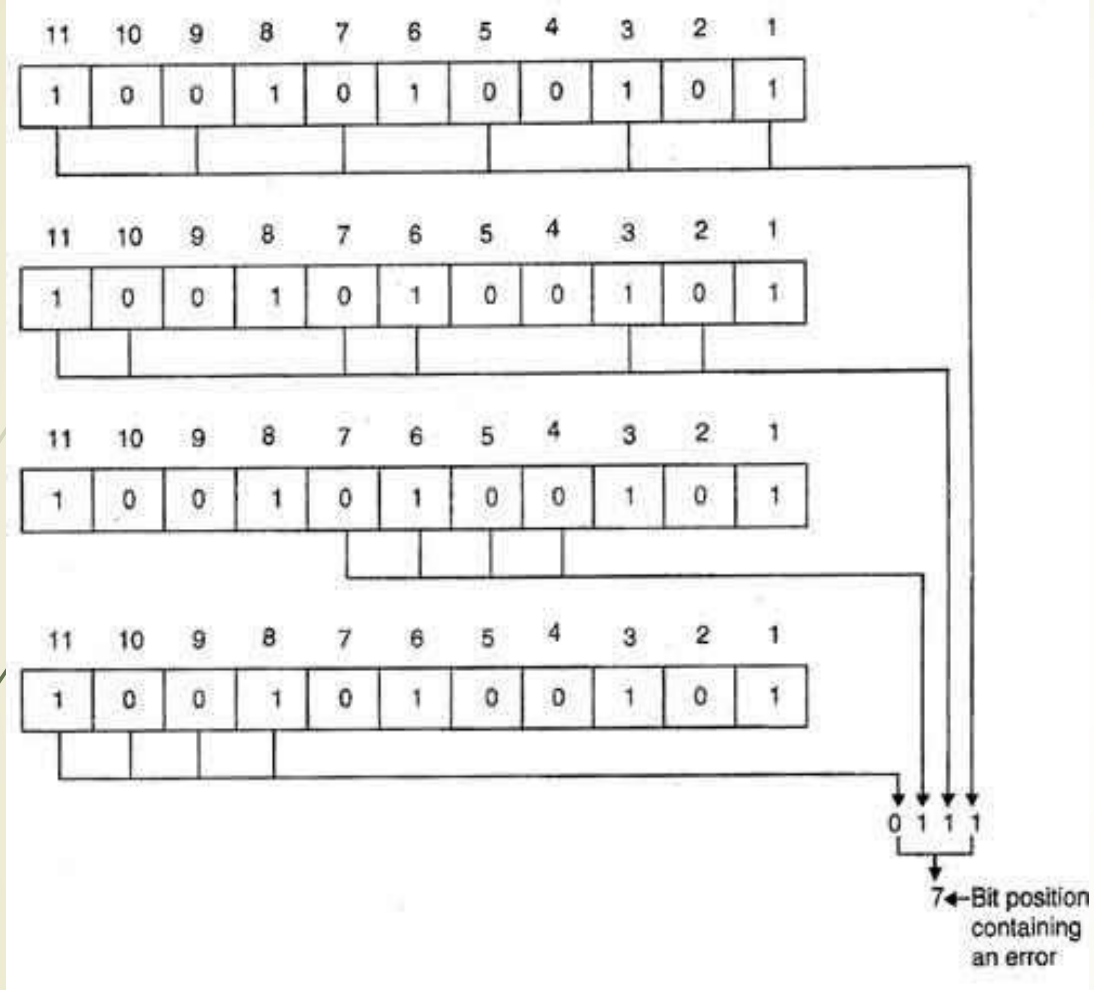


4. Thus data 1 0 0 1 1 1 0 0 1 0 1 will be transmitted.

Error Detection & Correction

Considering a case of above discussed example, if bit number 7 has been changed from 1 to 0.The data will be erroneous.

ERROR DETECTION AND CORRECTION



Data sent: 1 0 0 1 1 1 0 0 1 0 1

Data received: 1 00 1 0 1 00 1 0 1 (seventh bit changed)

HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

HDLC – A Brief Overview

- High Level Data Link Control is a protocol, which operates at the data link layer. The HDLC protocol embeds information in a data frame that allows devices to control data flow and correct errors
- HDLC is an ISO Standard developed from the Synchronous Data Link Control (SDLC) standard proposed by IBM
- Operates at the data link layer
- Used on both point-to-point and multipoint (multi-drop) data links
- Role of HDLC is to ensure that the data has been received without any loss or errors and in the correct order
- Provides connection-oriented and connection-less service
- ISO Standards: 3009, 4305

HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

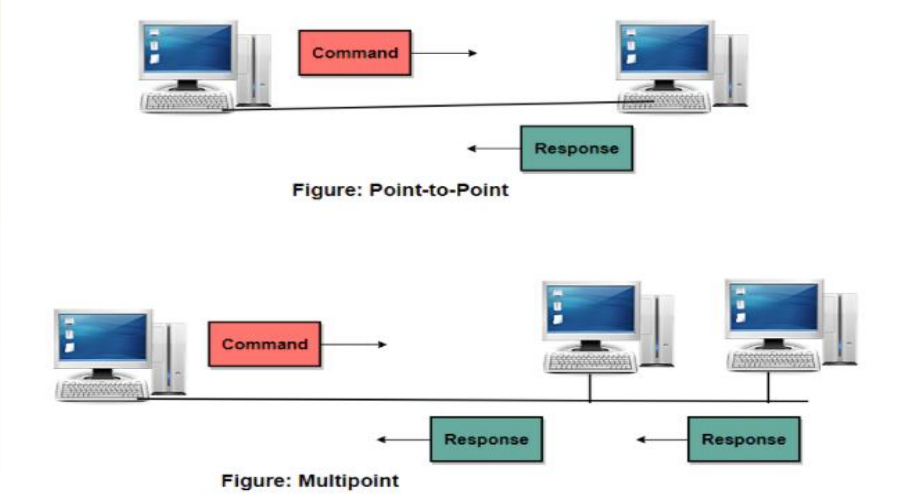
HDLC Basics:

- Stations:
  - Primary: sends data, controls the link with commands
  - Secondary: receives data, responds to control messages
  - Combined: can issue both commands and responses
- Link configuration:
  - Unbalanced: one primary station, one or more secondary stations
  - Balanced: two combined stations

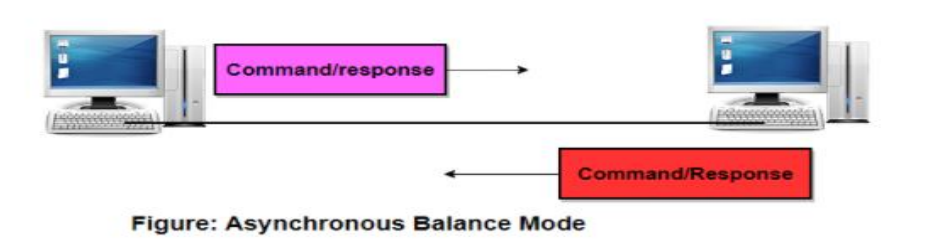
HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

Operation Modes

- HDLC has three operation modes
  - **Normal Response Mode (NRM)**
    - Used with unbalanced configuration
    - Primary initiates data transfer; secondary can only reply
  - **Asynchronous Response Mode (ARM)**
    - Secondary station initiates a transmission without receiving permission from the primary station
    - Primary terminal still retains responsibility for line initialization, error recovery, and logical disconnect
    - Allows the secondary station to send frames asynchronously with respect to the primary station
  - **Asynchronous Balanced Mode (ABM)**
    - Used with Balanced configuration
    - Either station may initiate the transmission at any time



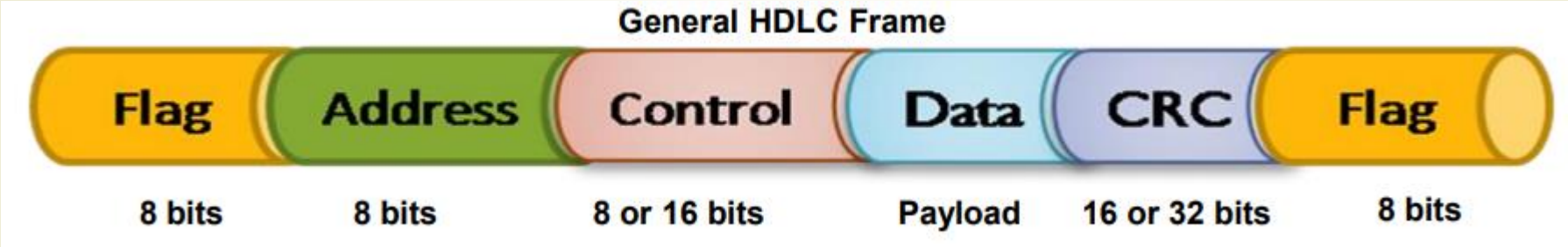
Normal Response Mode



Asynchronous Balanced Mode

HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

HDLC Frame Structure

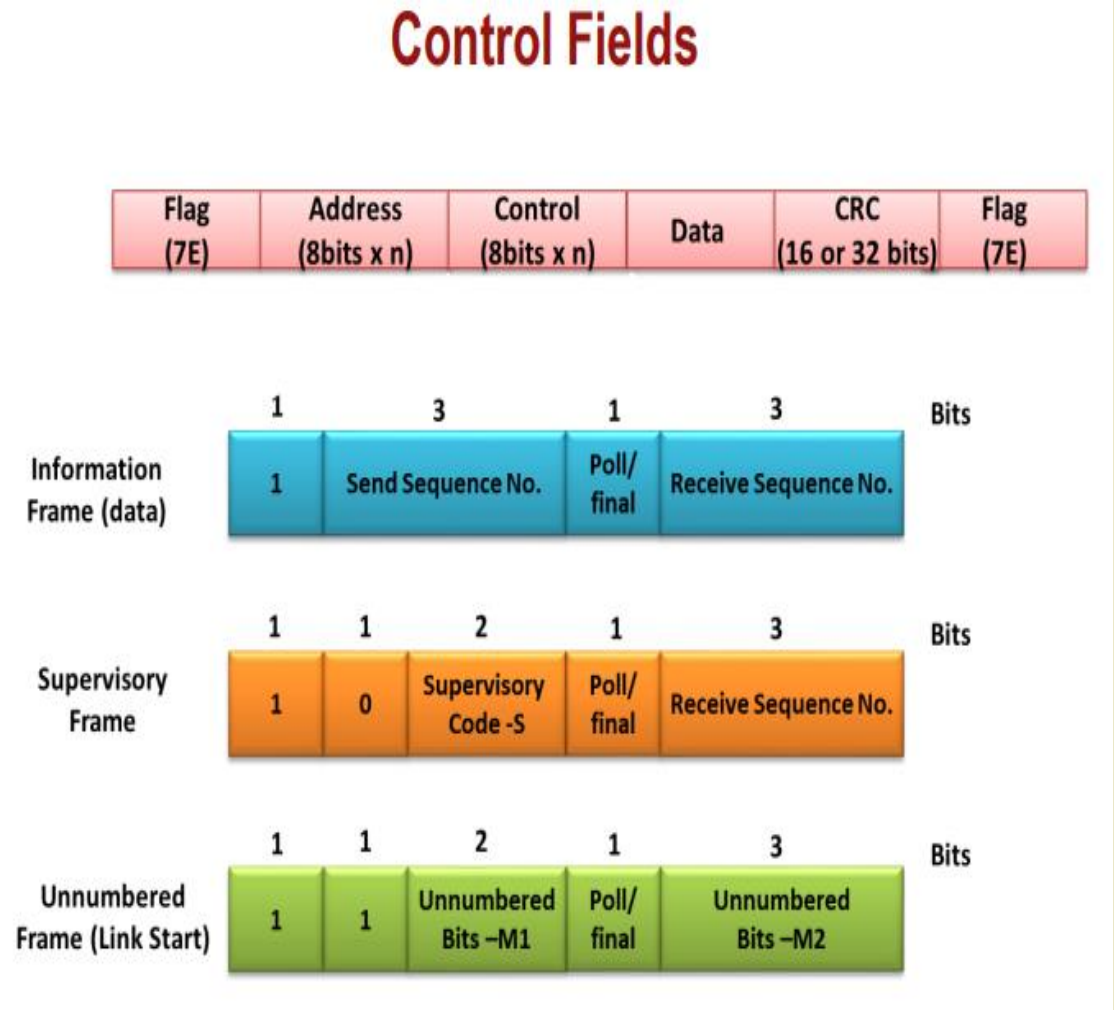


- **Flag** – Identifies the beginning and end of a frame -– 01111110 (7E Hex)
- **Address** – Address of the station: Single byte
- **Control** – Defines the frame type and is protocol dependent
- **Data** – Data field may vary in length depending upon the protocol using the frame. Layer 3 frames are carried in the data field
- **FCS** – Frame Check Sequence is used to verify the data integrity

HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

Frame Types:

- Three classes of frames are used
  - **Information frames (I-Frames)** – Carry the actual data. Transport user data from the network layer. In addition, they can also include flow and error control information piggybacked on data
  - **Supervisory frames (S-Frames)** – Used for error and flow control. They contain, send and receive sequence numbers
  - **Unnumbered frames (U-Frames)** – Used for various miscellaneous purposes, including link management



HIGH LEVEL DATA LINK CONTROL (HDLC) PROTOCOL

Information Frames (I-Frames)

1	3	1	3
1	Send Sequence No.	Poll/ final	Receive Sequence No.

- N(S): Sending Sequence Number
- N(R): Receiving sequence number
- P/F: Poll or Final bit

Supervisory Frames (S-Frames)

1	1	2	1	3
1	0	Supervisory Code - S	Poll/ final	Receive Sequence No.

- S =00 RR - Receiver Ready to accept more I-frames (data)
- S =10 RNR - Receiver Not Ready to accept more I-frames
- S =01 REJ - Go-Back-N retransmission request for an I-frame
- S =11 SREJ - Selective retransmission request for an I-frame

Unnumbered Frames (U-Frames)

1	1	2	1	3
1	1	M1	Poll/ final	M2

- SNRM: set normal response mode (M1 = 00, M2 = 001)
- SABM: set asynchronous balanced mode (M1 = 11, M2 = 100)
- SABME: set asynchronous balanced mode, extended (M1 = 11, M2 = 110)
- DISC: disconnect (M1=00, M2=010)
- UA: un-numbered acknowledgement (M1 = 00, M2 = 110)
- RSET: resets send and receive sequence numbers (M1 = 11, M2=001)
- FRMR: frame reject (M1 = 10, M2=001)

POINT – TO – POINT PROTOCOL

- Point - to - Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit data between two directly connected (point-to-point) computers.
- It is a byte - oriented protocol that is widely used in broadband communications having heavy loads and high speeds.
- Since it is a data link layer protocol, data is transmitted in frames.
- It is also known as RFC 1661.

Services Provided by PPP

- The main services provided by Point - to - Point Protocol are −
  - Defining the frame format of the data to be transmitted.
  - Defining the procedure of establishing link between two points and exchange of data.
  - Stating authentication rules of the communicating devices.
  - Providing address for network communication.

POINT – TO – POINT PROTOCOL

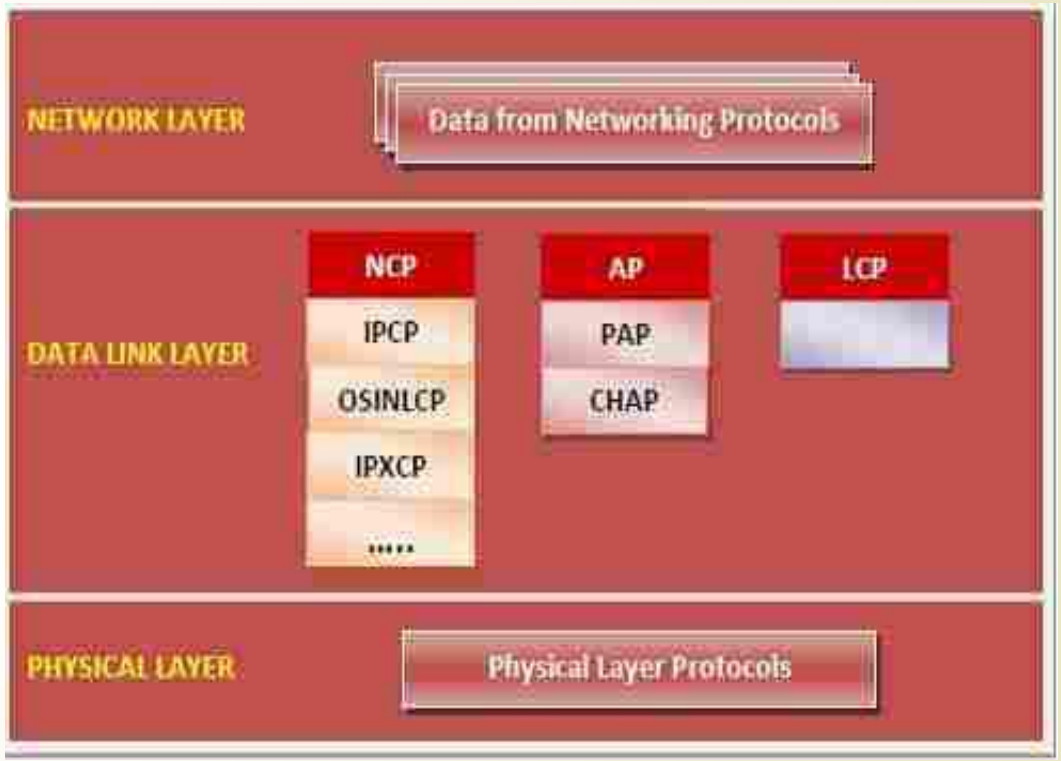
- Providing connections over multiple links.
- Supporting a variety of network layer protocols by providing a range of services.

Components of PPP

- Point - to - Point Protocol is a layered protocol having different components −
- Encapsulation Component − It encapsulates the datagram so that it can be transmitted over the specified physical layer.
  - **Link Control Protocol (LCP)** − It is responsible for establishing, configuring, testing, maintaining and terminating links for transmission. It also imparts negotiation for set up of options and use of features by the two endpoints of the links.
  - **Authentication Protocols (AP)** − These protocols authenticate endpoints for use of services. The two authentication protocols of PPP are −
    - Password Authentication Protocol (PAP)
    - Challenge Handshake Authentication Protocol (CHAP)

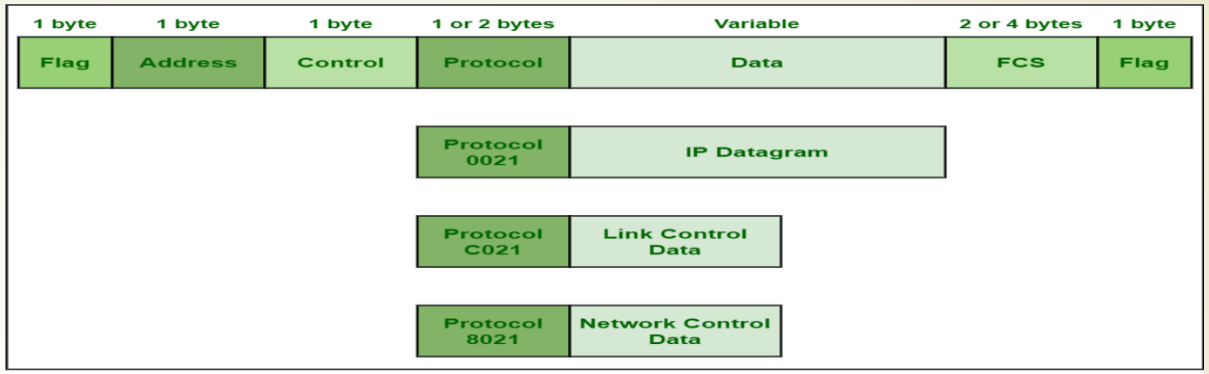
POINT – TO – POINT PROTOCOL

- **Network Control Protocols (NCPs)** – These protocols are used for negotiating the parameters and facilities for the network layer. For every higher-layer protocol supported by PPP, one NCP is there. Some of the NCPs of PPP are –
  - Internet Protocol Control Protocol (IPCP)
  - OSI Network Layer Control Protocol (OSINLCP)
  - Internetwork Packet Exchange Control Protocol (IPXCP)
  - DECnet Phase IV Control Protocol (DNCP)
  - NetBIOS Frames Control Protocol (NBFCP)
  - IPv6 Control Protocol (IPV6CP)



POINT – TO – POINT PROTOCOL

PPP Frame



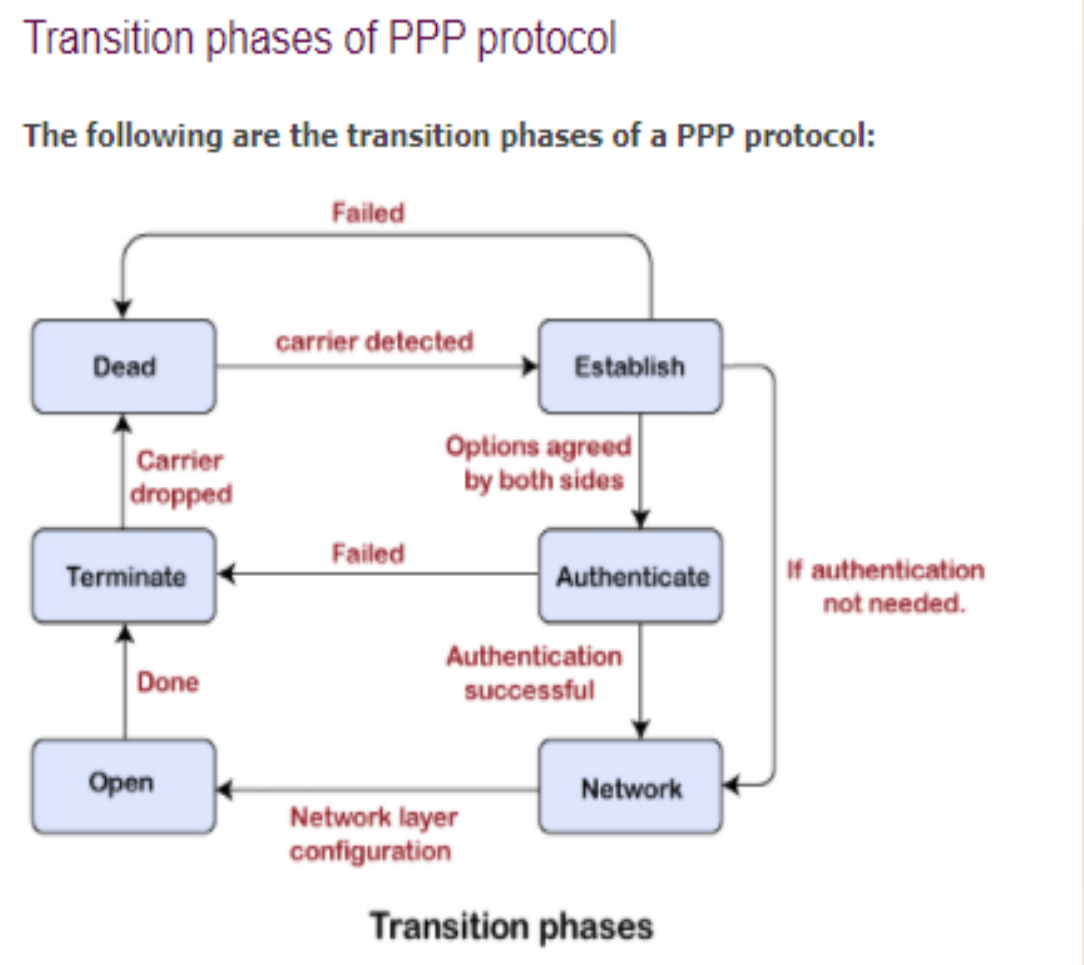
PPP Frame Format

- PPP is a byte - oriented protocol where each field of the frame is composed of one or more bytes. The fields of a PPP frame are −
  - **Flag** − 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
  - **Address** − 1 byte which is set to 11111111 in case of broadcast.
  - **Control** − 1 byte set to a constant value of 00000011.
  - **Protocol** − 1 or 2 bytes that define the type of data contained in the payload field.
  - **Payload** − This carries the data from the network layer. The maximum length of the payload field is 1500 bytes. However, this may be negotiated between the endpoints of communication.
  - **FCS** − It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

POINT – TO – POINT PROTOCOL

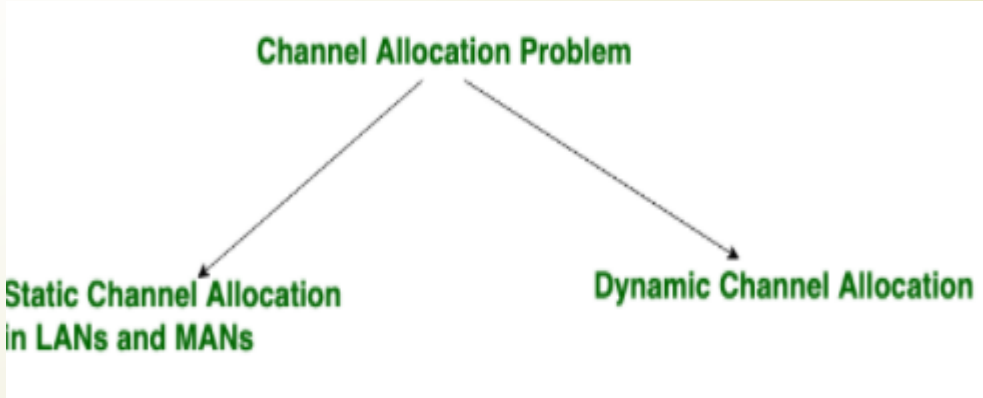
Byte Stuffing in PPP Frame

- Byte stuffing is used in PPP payload field whenever the flag sequence appears in the message, so that the receiver does not consider it as the end of the frame.
- The escape byte, 01111101, is stuffed before every byte that contains the same byte as the flag byte or the escape byte.
- The receiver on receiving the message removes the escape byte before passing it onto the network layer.



CHANNEL ALLOCATION PROBLEM

- Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks.
- There are user's quantity may vary every time the process takes place.
- If there are N number of users and channel is divided into N equal-sized sub channels, Each user is assigned one portion.
- If the number of users are small and don't vary at times, than Frequency Division Multiplexing can be used as it is a simple and efficient channel bandwidth allocating technique.
- Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.



CHANNEL ALLOCATION PROBLEM

Static Channel Allocation in LANs and MANs:

- It is the classical or traditional approach of allocating a single channel among multiple competing users Frequency Division Multiplexing (FDM).
- If there are N users, the bandwidth is divided into N equal sized portions each user being assigned one portion.
- since each user has a private frequency band, there is no interface between users.
- It is not efficient to divide into fixed number of chunks.

Dynamic Channel Allocation

- In dynamic channel allocation scheme, frequency bands are not permanently assigned to the users. Instead channels are allotted to users dynamically as needed, from a central pool.
- The allocation is done considering a number of parameters so that transmission interference is minimized

CHANNEL ALLOCATION PROBLEM

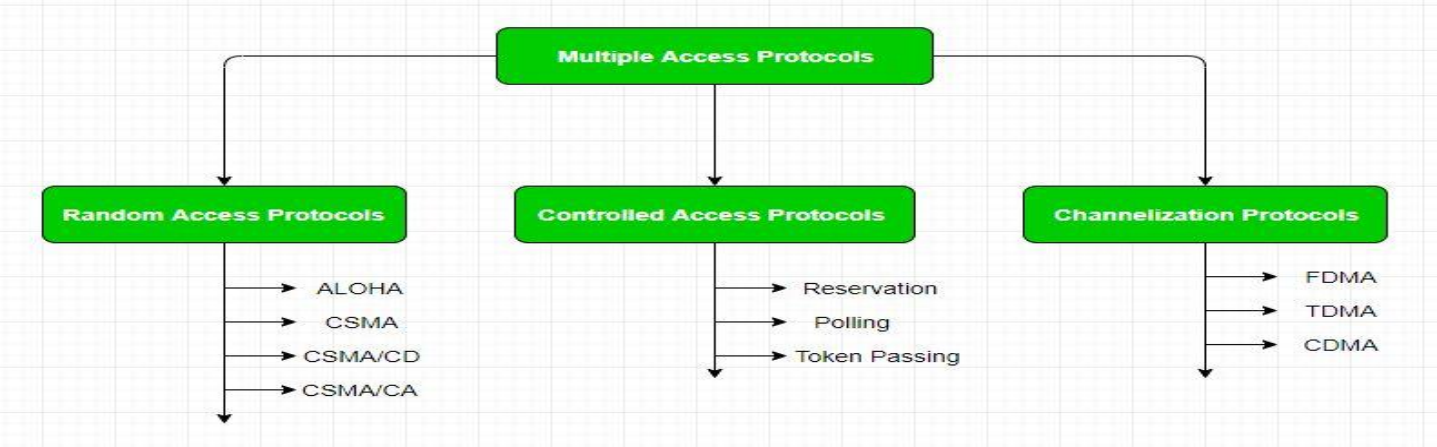
- This allocation scheme optimizes bandwidth usage and results is faster transmissions.
- Dynamic channel allocation is further divided into centralized and distributed allocation.

Possible assumptions include:

- **Station Model:**
  - Assumes that each of N stations independently produce frames. The probability of producing a packet in the interval IDt where I is the constant arrival rate of new frames.
- **Single Channel Assumption:**
  - In this allocation all stations are equivalent and can send and receive on that channel.
- **Collision Assumption:**
  - If two frames overlap in time-wise, then that's collision. Any collision is an error, and both frames must re transmitted. Collisions are only possible error.
- **Time**
  - can be divided into Slotted or Continuous.
- **Stations**
  - can sense a channel is busy before they try it.

MULTIPLE ACCESS

- If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously.
- Hence multiple access protocols are required to decrease collision and avoid crosstalk.
- For example, in a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of chaos is created(data overlap or data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.
- Thus, protocols are required for sharing data on non dedicated channels. Multiple access protocols can be subdivided further as –



MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

- In this, all stations have same superiority that is no station has more priority than another station.
- Any station can send data depending on medium's state(idle or busy). It has two features:
  - There is no fixed time for sending data
  - There is no fixed sequence of stations sending data

ALOHA (an Hawaiian word, meaning "hello".)

- Aloha is a multiple access protocol at the data link layer and proposes how multiple terminals access the medium without interference or collision.
- The Aloha protocol was designed as part of a project at the University of Hawaii.
- It provided data transmission between computers on several of the Hawaiian Islands involving packet radio networks.
- **There are two different versions of ALOHA:**

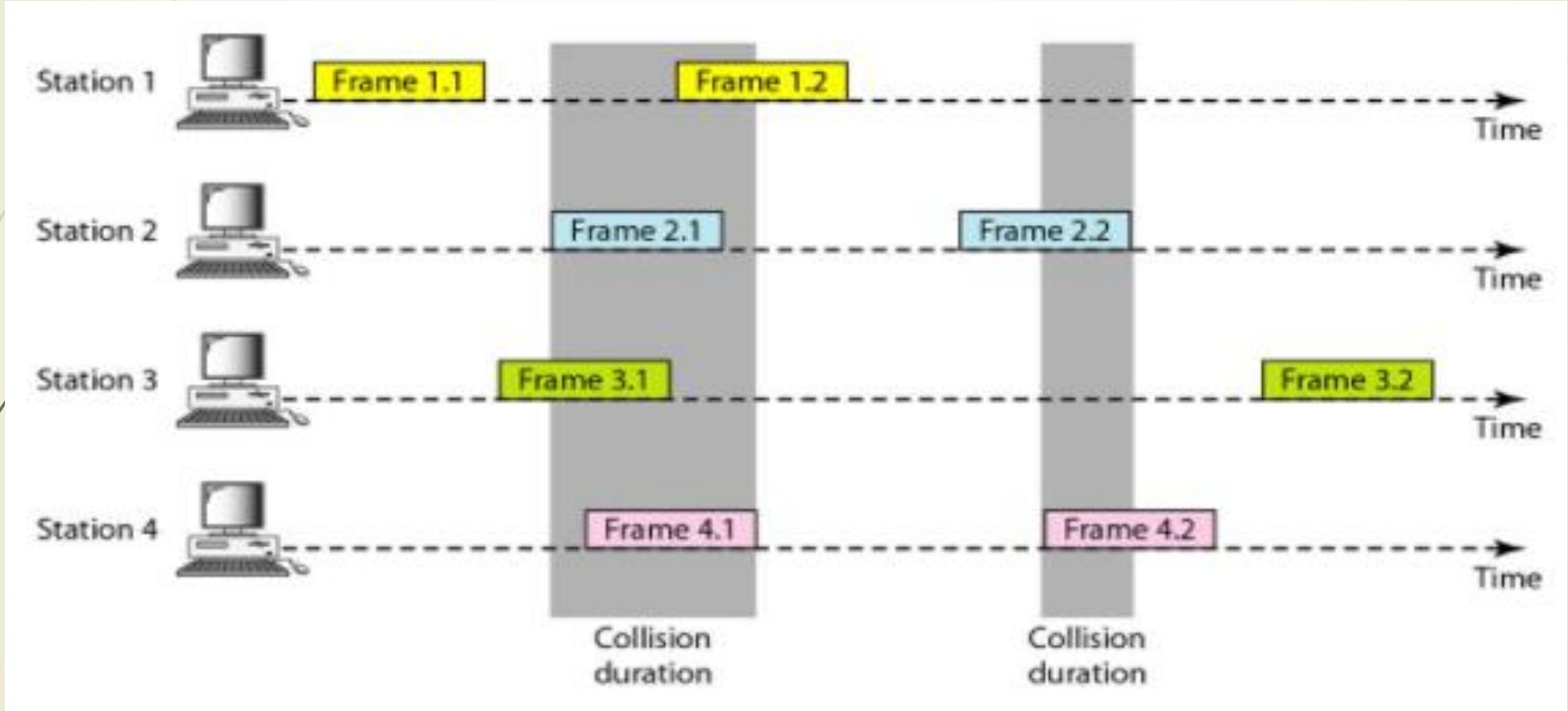
MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

Pure ALOHA

- In pure ALOHA, the stations transmit frames whenever they have data to send.
- When two or more stations transmit simultaneously, there is collision and the frames are destroyed.
- In pure ALOHA, whenever any station transmits a frame, it expects the acknowledgement from the receiver.
- If acknowledgement is not received within specified time, the station assumes that the frame (or acknowledgement) has been destroyed.
- If the frame is destroyed because of collision the station waits for a random amount of time and sends it again. This waiting time must be random otherwise same frames will collide again and again.
- Therefore pure ALOHA dictates that when time-out period passes, each station must wait for a random amount of time before re-sending its frame. This randomness will help avoid more collisions.
- Figure shows an example of frame collisions in pure ALOHA.

MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

Pure ALOHA



MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

- In fig there are four stations that contended with one another for access to shared channel.
- All these stations are transmitting frames.
- Some of these frames collide because multiple frames are in contention for the shared channel.
- Only two frames, frame 1.1 and frame 3.2 survive. All other frames are destroyed.
- Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be damaged.
- If first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted.

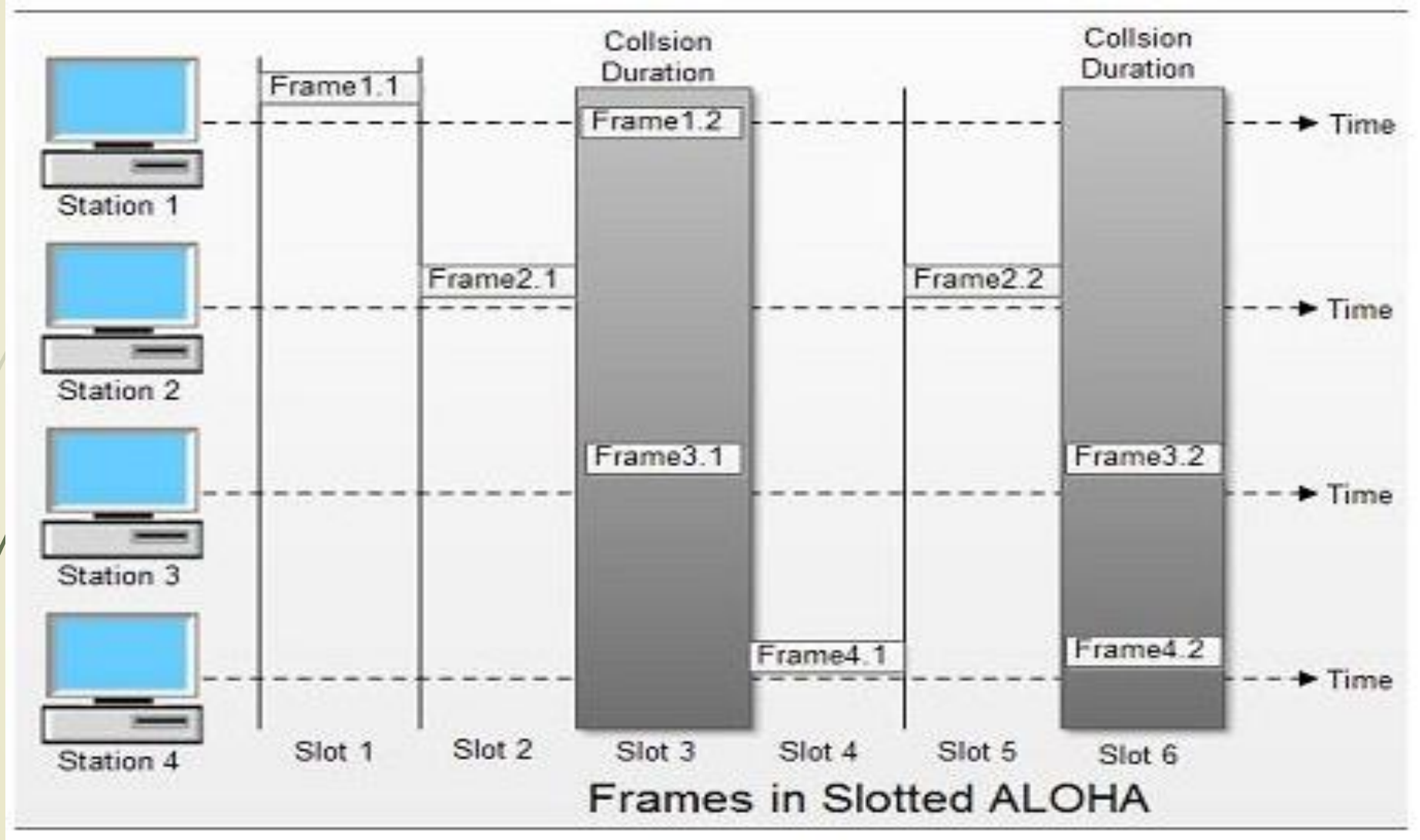
MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

Slotted ALOHA

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA as chances of collision in pure ALOHA are very high.
- In slotted ALOHA, the time of the shared channel is divided into discrete intervals called slots.
- The stations can send a frame only at the beginning of the slot and only one frame is sent in each slot.
- In slotted ALOHA, if any station is not able to place the frame onto the channel at the beginning of the slot i.e. it misses the time slot then the station has to wait until the beginning of the next time slot.
- In slotted ALOHA, there is still a possibility of collision if two stations try to send at the beginning of the same time slot as shown in fig.
- Slotted ALOHA still has an edge over pure ALOHA as chances of collision are reduced to one-half.

MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

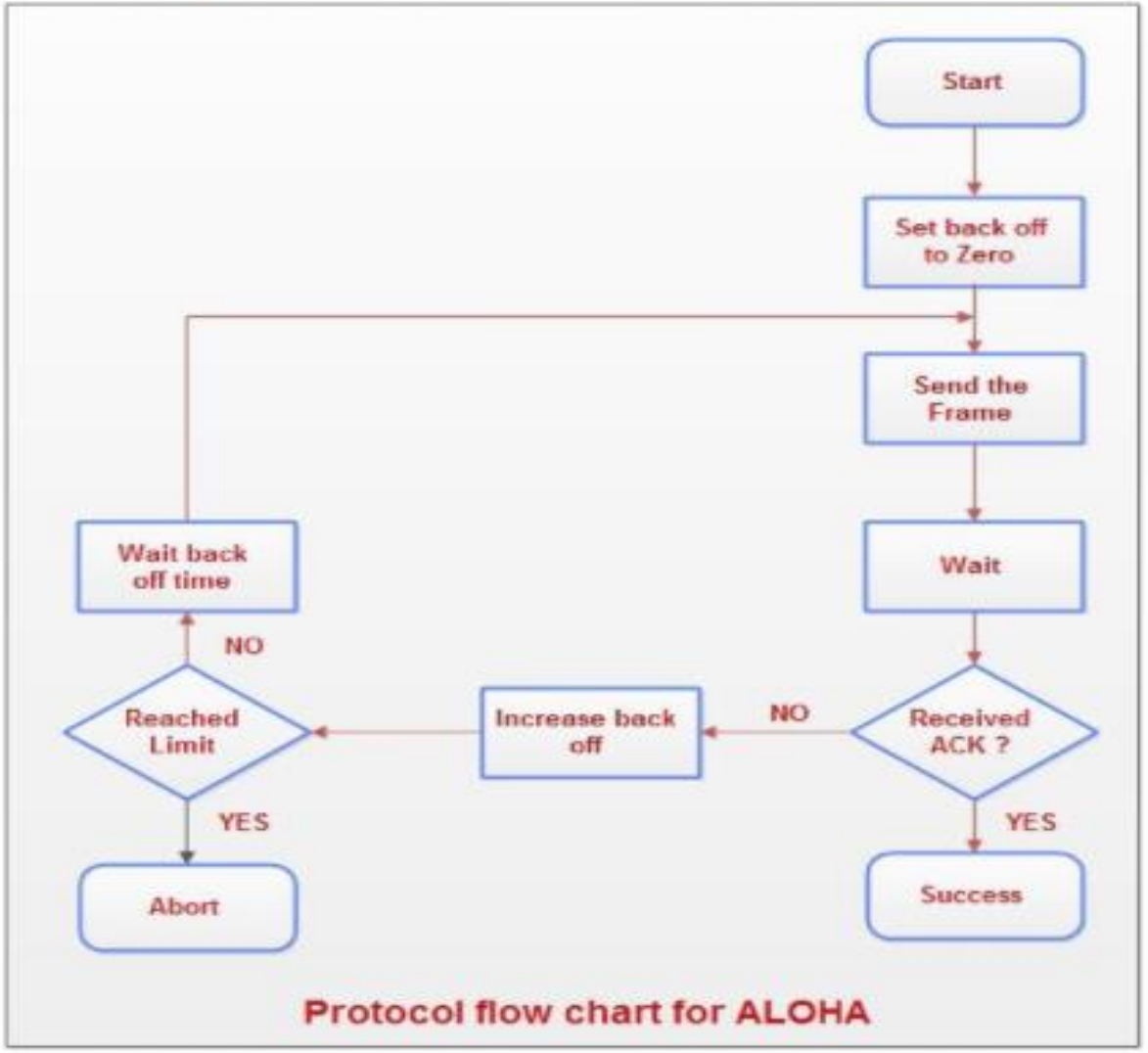
Slotted ALOHA



MULTIPLE ACCESS- RANDOM ACCESS - ALOHA

Protocol Flow Chart for ALOHA

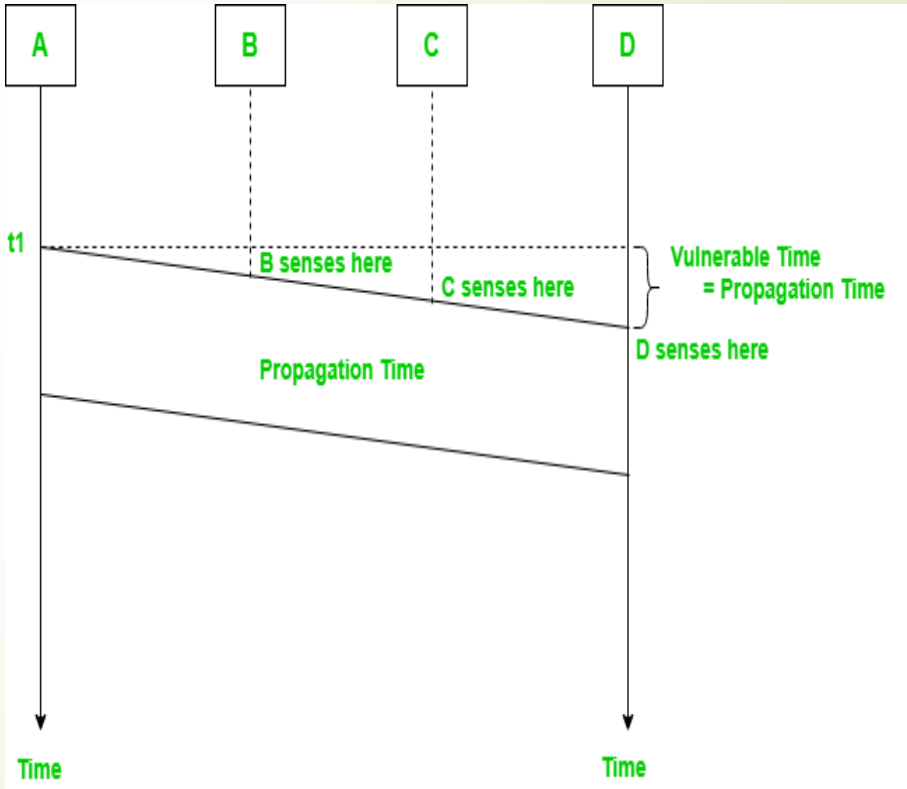
Fig. shows the protocol flow chart for ALOHA.



Protocol flow chart for ALOHA

MULTIPLE ACCESS- RANDOM ACCESS -CSMA

- This method was developed to decrease the chances of collisions when two or more stations start sending their signals over the datalink layer.
- **Carrier Sense multiple access** requires that each station first check the state of the medium before sending.
- It is a **carrier sense multiple access** based on media access protocol to sense the traffic on a channel (idle or busy) before transmitting the data.
- It means that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle.
- Hence, it reduces the chances of a collision on a transmission medium.
- The persistence methods can be applied to help the station take action when the channel is busy/idle.

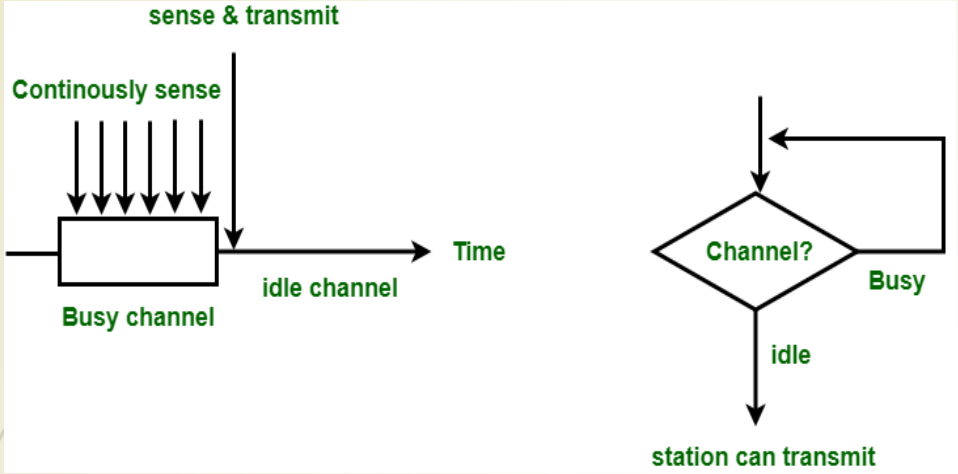


MULTIPLE ACCESS- RANDOM ACCESS -CSMA

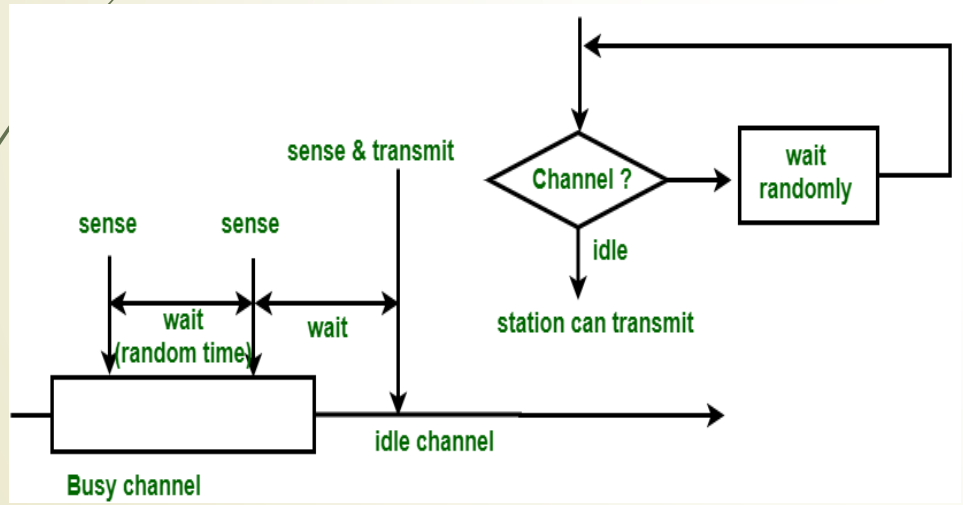
CSMA Access Modes

- **1-Persistent:**
  - In the 1-Persistent mode of CSMA that defines each node, first sense the shared channel and if the channel is idle, it immediately sends the data.
  - Else it must wait and keep track of the status of the channel to be idle and broadcast the frame unconditionally as soon as the channel is idle.
- **Non-Persistent:**
  - It is the access mode of CSMA that defines before transmitting the data, each node must sense the channel, and if the channel is inactive, it immediately sends the data.
  - Otherwise, the station must wait for a random time (not continuously), and when the channel is found to be idle, it transmits the frames.
- **P-Persistent:**
  - It is the combination of 1-Persistent and Non-persistent modes.
  - The P-Persistent mode defines that each node senses the channel, and if the channel is inactive, it sends a frame with a P probability.
  - If the data is not transmitted, it waits for a (q = 1-p probability) random time and resumes the frame with the next time slot.

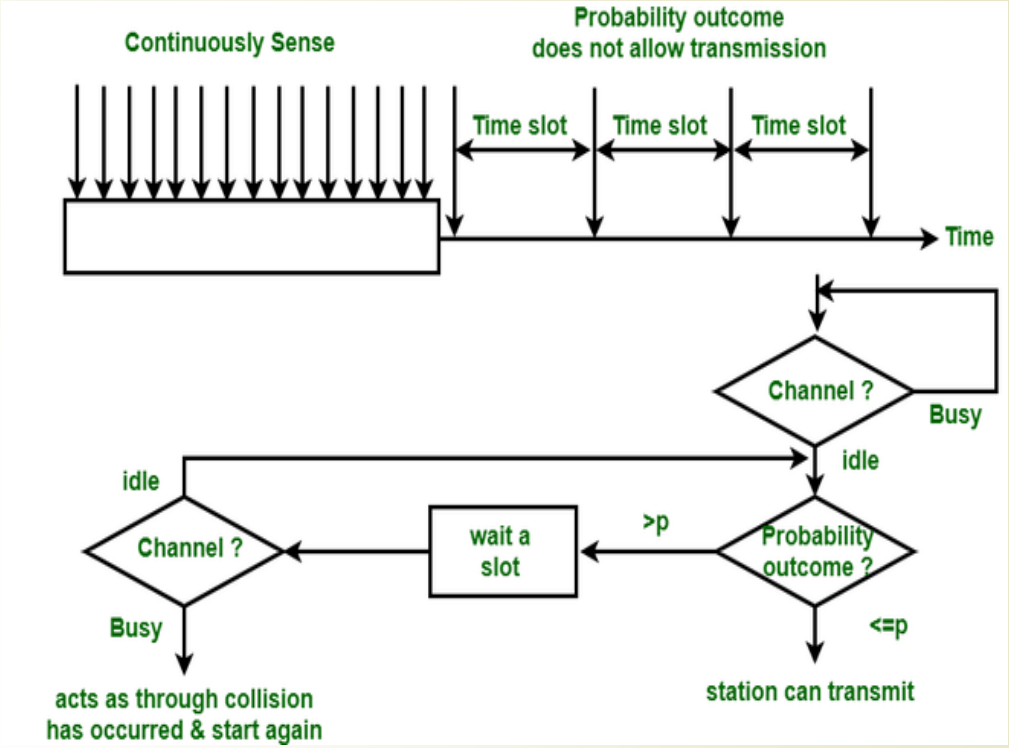
MULTIPLE ACCESS- RANDOM ACCESS -CSMA



1-Persistence



Non-Persistence



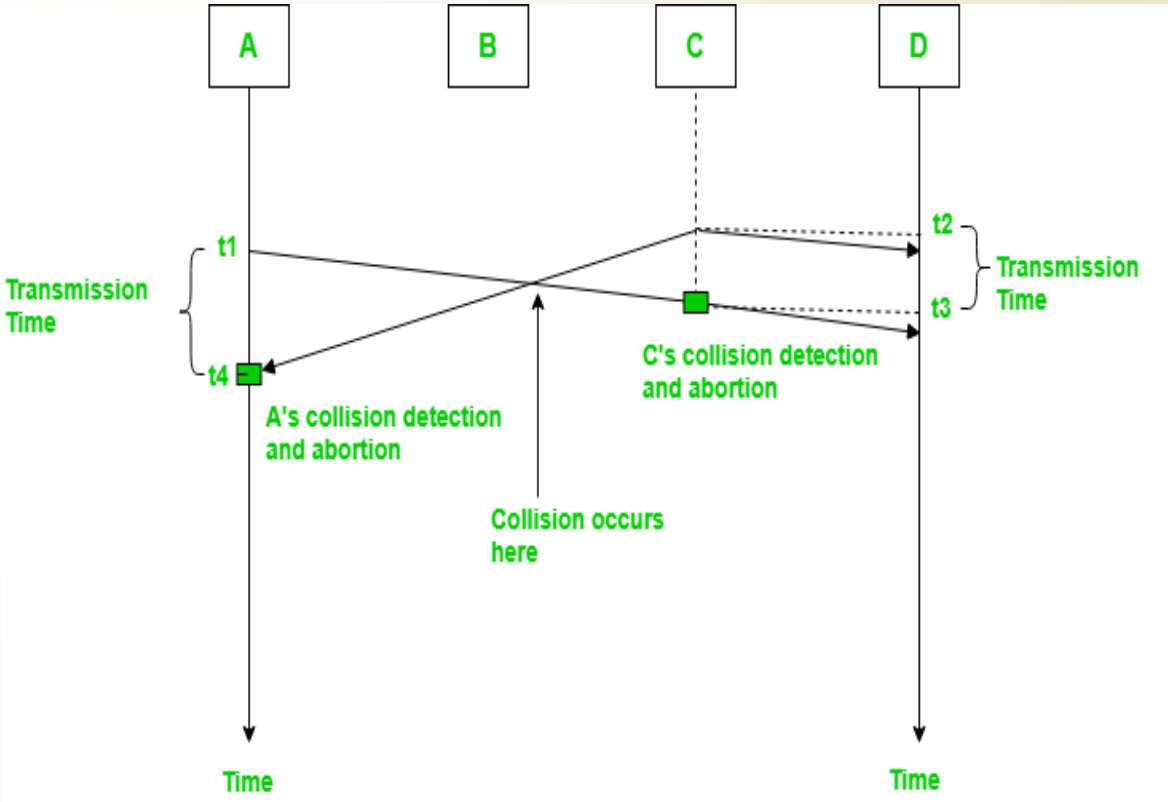
P-Persistence

MULTIPLE ACCESS- RANDOM ACCESS –CSMA/CD

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If successful, the station is finished, if not, the frame is sent again.

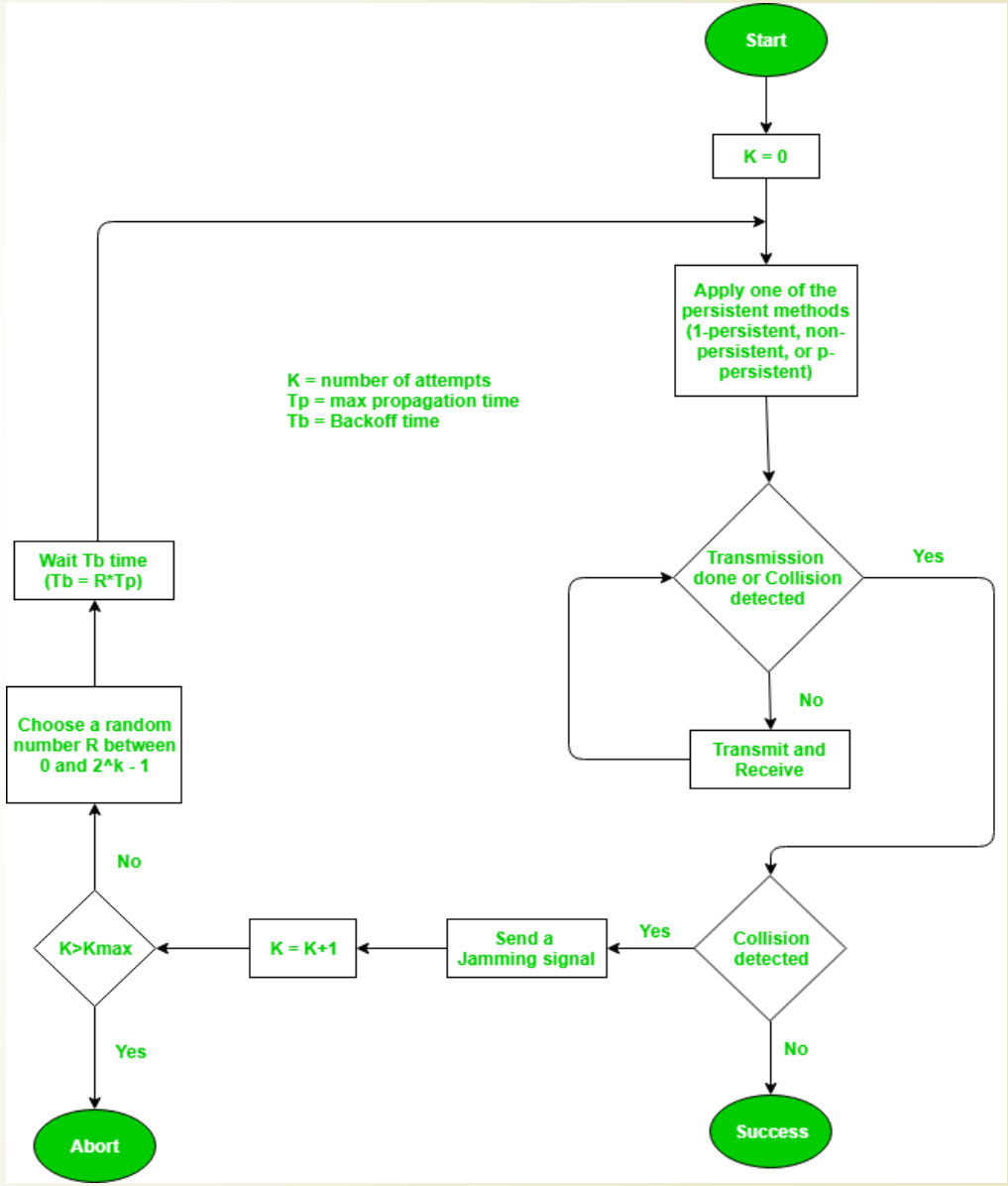
- In the diagram, A starts send the first bit of its frame at t1 and since C sees the channel idle at t2, starts sending its frame at t2.
- C detects A's frame at t3 and aborts transmission. A detects C's frame at t4 and aborts its transmission.
- Transmission time for C's frame is therefore t3-t2 and for A's frame is t4-t1.



MULTIPLE ACCESS- RANDOM ACCESS –CSMA/CD

So, the frame transmission time (Tfr) should be at least twice the maximum propagation time (Tp).

- This can be deduced when the two stations involved in collision are maximum distance apart.
- Throughput and Efficiency –
- The throughput of CSMA/CD is much greater than pure or slotted ALOHA.
  - For 1-persistent method throughput is 50%
  - For non-persistent method throughput can go up to 90%.



MULTIPLE ACCESS- RANDOM ACCESS –CSMA/CA

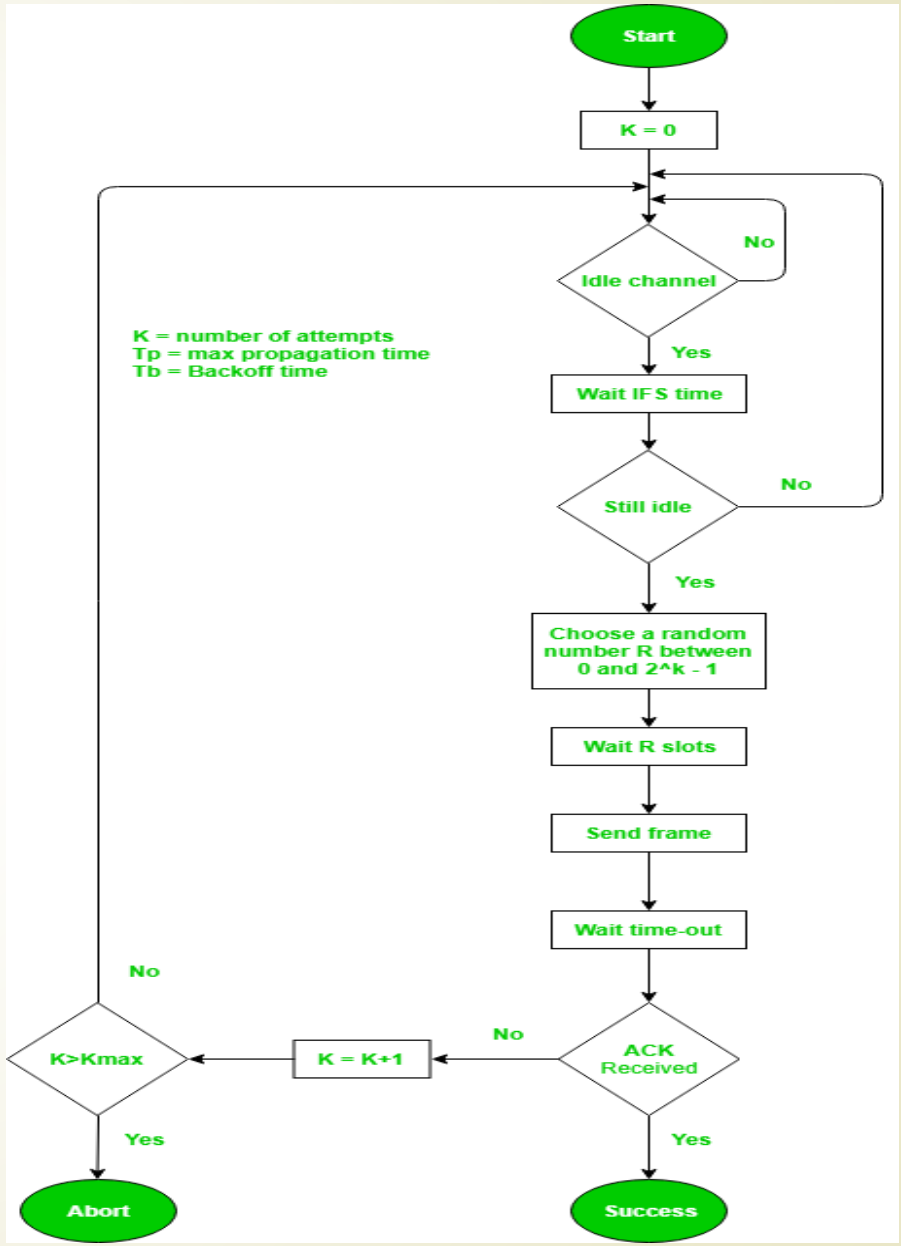
Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer.
- In contrast to CSMA/CD (Carrier Sense Multiple Access/Collision Detection) that deals with collisions after their occurrence, CSMA/CA prevents collisions prior to their occurrence
- In wired networks, if a collision has occurred then the energy of received signal almost doubles and the station can sense the possibility of collision.
- In case of wireless networks, most of the energy is used for transmission and the energy of received signal increases by only 5-10% if a collision occurs.
- It can't be used by the station to sense collision. Therefore CSMA/CA has been specially designed for wireless networks.

MULTIPLE ACCESS- RANDOM ACCESS –CSMA/CA

These are three types of strategies:

- **InterFrame Space (IFS) –**
  - When a station finds the channel busy, it waits for a period of time called IFS time. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority
- **Contention Window –**
  - It is the amount of time divided into slots. A station which is ready to send frames chooses random number of slots as wait time.
- **Acknowledgements –**
  - The positive acknowledgements and time-out timer can help guarantee a successful transmission of the frame.



MULTIPLE ACCESS- RANDOM ACCESS –CSMA/CA

S.NO	CSMA/CD	CSMA/CA
1.	CSMA / CD is effective after a collision.	Whereas CSMA / CA is effective before a collision.
2.	CSMA / CD is used in wired networks.	Whereas CSMA / CA is commonly used in wireless networks.
3.	It only reduces the recovery time.	Whereas CSMA/ CA minimizes the possibility of collision.
4.	CSMA / CD resend the data frame whenever a conflict occurs.	Whereas CSMA / CA will first transmit the intent to send for data transmission.
5.	CSMA / CD is used in 802.3 standard.	While CSMA / CA is used in 802.11 standard.
6.	It is more efficient than simple CSMA(Carrier Sense Multiple Access).	While it is similar to simple CSMA(Carrier Sense Multiple Access).

CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

- In controlled access, the stations seek information from one another to find which station has the right to send.
- It allows only one node to send at a time, to avoid collision of messages on shared medium.
- The three controlled-access methods are:
  - Reservation
  - Polling
  - Token Passing

Reservation

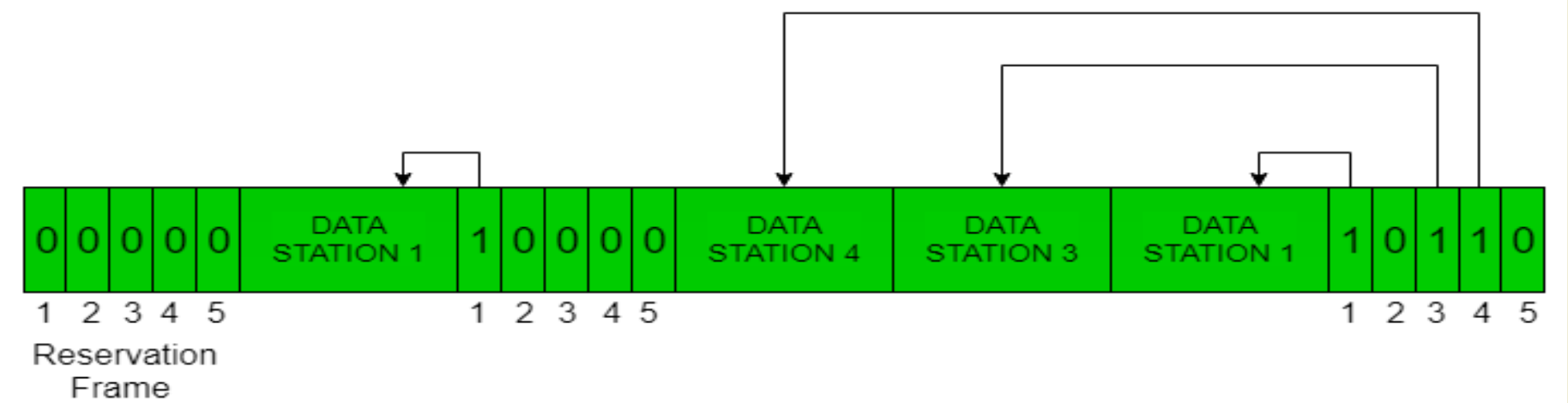
- In the reservation method, a station needs to make a reservation before sending data.
- The time line has two kinds of periods:
  - Reservation interval of fixed time length
  - Data transmission period of variable frames.

CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

- If there are M stations, the reservation interval is divided into M slots, and each station has one slot.
- Suppose if station 1 has a frame to send, it transmits 1 bit during the slot 1. No other station is allowed to transmit during this slot.
- In general, i^{th} station may announce that it has a frame to send by inserting a 1 bit into i^{th} slot.
- After all N slots have been checked, each station knows which stations wish to transmit.
- The stations which have reserved their slots transfer their frames in that order.
- After data transmission period, next reservation interval begins.
- Since everyone agrees on who goes next, there will never be any collisions.s

CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

The following figure shows a situation with five stations and a five slot reservation frame. In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.

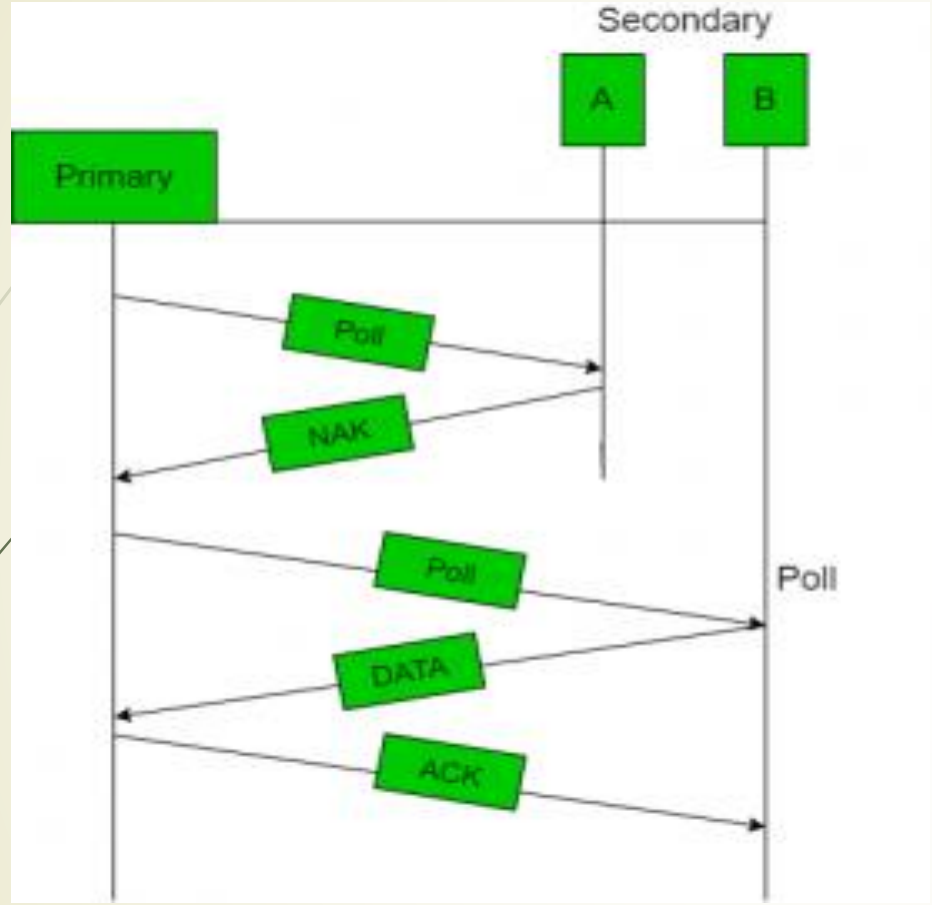


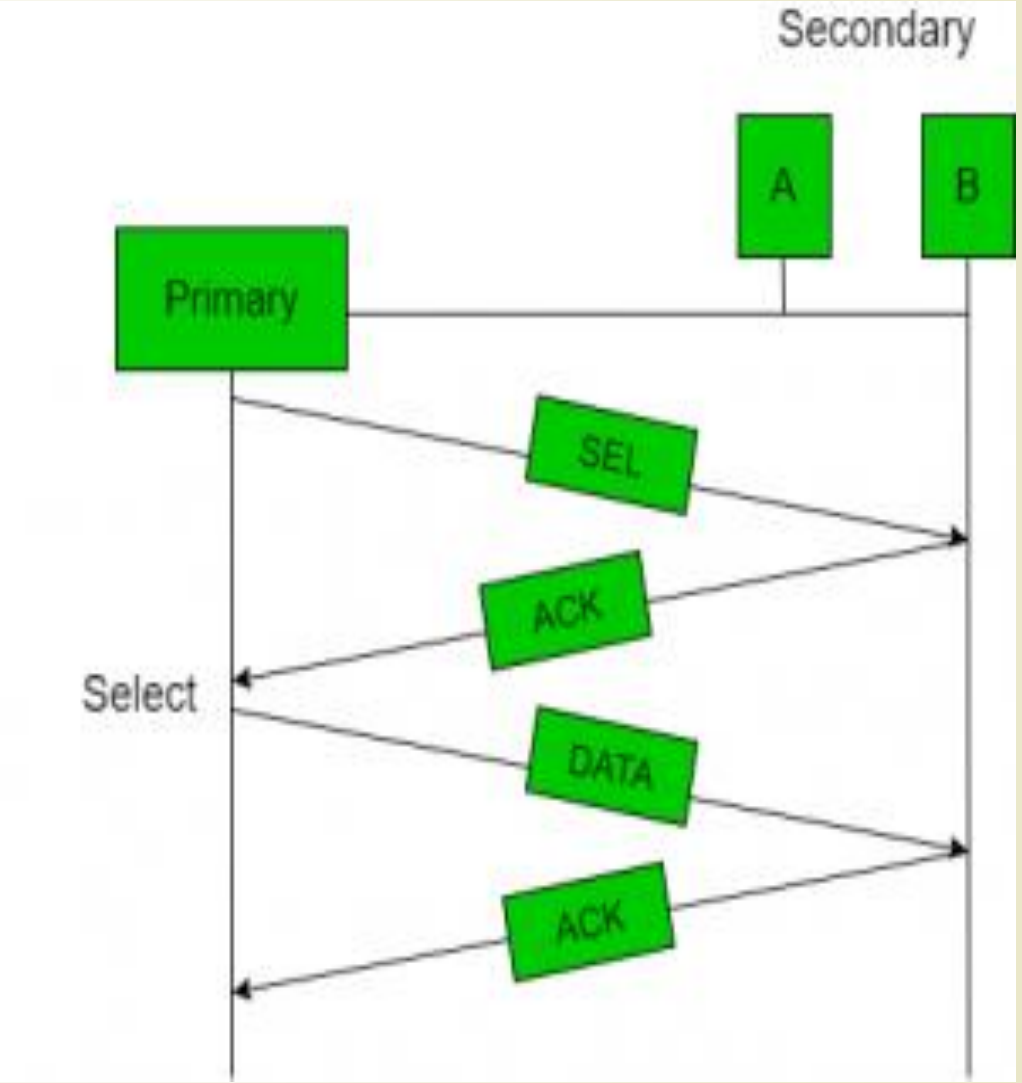
CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

Polling

- Polling process is similar to the roll-call performed in class. Just like the teacher, a controller sends a message to each node in turn.
- In this, one acts as a primary station(controller) and the others are secondary stations. All data exchanges must be made through the controller.
- The message sent by the controller contains the address of the node being selected for granting access.
- Although all nodes receive the message but the addressed one responds to it and sends data, if any. If there is no data, usually a “poll reject”(NAK) message is sent back.
- Problems include high overhead of the polling messages and high dependence on the reliability of the controller.
- **Efficiency**
- Let T_{poll} be the time for polling and T_t be the time required for transmission of data. Then,
- Efficiency = $T_t/(T_t + T_{poll})$

CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK





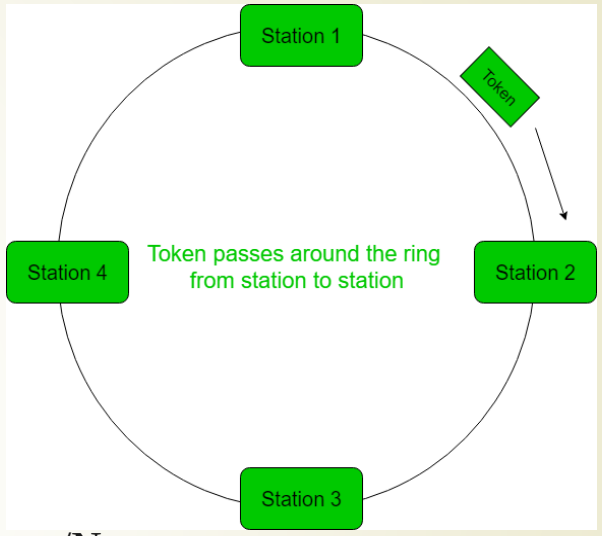
CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

Token Passing

- In token passing scheme, the stations are connected logically to each other in form of ring and access of stations is governed by tokens.
- A token is a special bit pattern or a small message, which circulate from one station to the next in the some predefined order.
- In Token ring, token is passed from one station to another adjacent station in the ring whereas incase of Token bus, each station uses the bus to send the token to the next station in some predefined order.
- In both cases, token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station. If it has no queued frame, it passes the token simply.
- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbors and the other N – 1 stations to send a frame, if they have one.

CONTROLLED ACCESS PROTOCOLS IN COMPUTER NETWORK

- There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.



- **Performance**
  - Performance of token ring can be concluded by 2 parameters:-
  - **Delay**, which is a measure of time between when a packet is ready and when it is delivered.
  - So, the average time (delay) required to send a token to the next station = a/N.
  - **Throughput**, which is a measure of the successful traffic.

Throughput, $S = 1/(1 + a/N)$ for $a<1$

and

$$S = 1/\{a(1 + 1/N)\} \text{ for } a>1.$$

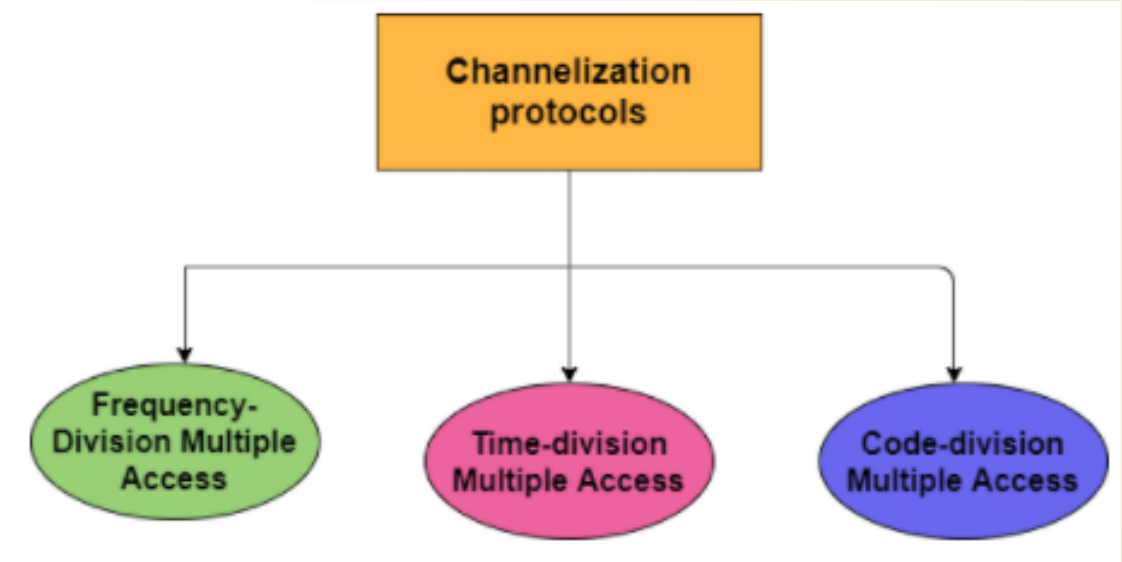
where N = number of stations

$a = T_p/T_t$

(T_p = propagation delay and T_t = transmission delay)

CHANNELIZATION

- Channelization is basically a method that provides the multiple-access and in this, the available bandwidth of the link is shared in time, frequency, or through the code in between the different stations.
- Channelization Protocols are broadly classified as follows:
  - FDMA(Frequency-Division Multiple Access)
  - TDMA(Time-Division Multiple Access)
  - CDMA(Code-Division Multiple Access)



CHANNELIZATION

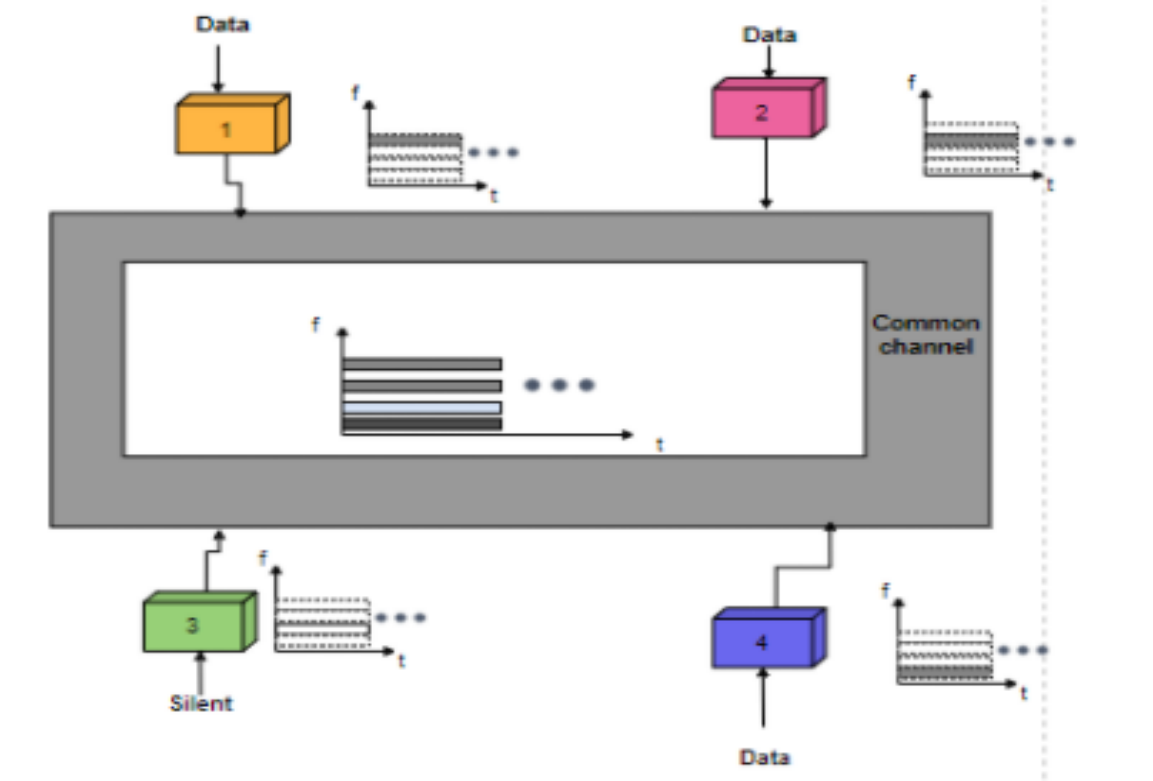
Frequency-Division Multiple Access

- With the help of this technique, the available bandwidth is divided into frequency bands.
- Each station is allocated a band in order to send its data. Or in other words, we can say that each band is reserved for a specific station and it belongs to the station all the time.
- Each station makes use of the bandpass filter in order to confine the frequencies of the transmitter.
- In order to prevent station interferences, the allocated bands are separated from one another with the help of small guard bands.
- The Frequency-division multiple access mainly specifies a predetermined frequency for the entire period of communication.
- Stream of data can be easily used with the help of FDMA.
- **Advantages of FDMA**
  - This technique is efficient when the traffic is uniformly constant
  - In case if the channel is not in use then it sits idle.

CHANNELIZATION

- FDMA is simple algorithmically and the complexity is less.
- For FDMA there is no restriction regarding the type of baseband or the type of modulation

- **Disadvantages of FDMA**
  - By using FDMA, the maximum flow rate per channel is fixed and small.



CHANNELIZATION

Time-Division Multiple Access

- Time-Division Multiple access is another method to access the channel for shared medium networks.
- With the help of this technique, the stations share the bandwidth of the channel in time.
- A time slot is allocated to each station during which it can send the data.
- Data is transmitted by each station in the assigned time slot.
- There is a problem in using TDMA and it is due to TDMA the synchronization cannot be achieved between the different stations.
- When using the TDMA technique then each station needs to know the beginning of its slot and the location of its slot.
- If the stations are spread over a large area, then there occur propagation delays; in order to compensate this guard, times are used.
- The data link layer in each station mainly tells its physical layer to use the allocated time slot.

CHANNELIZATION

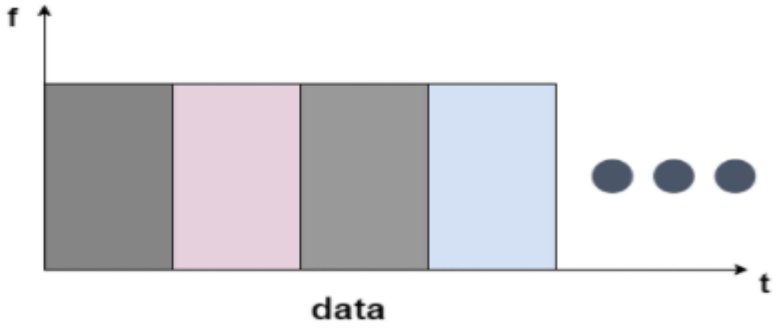


Figure: Time-Division media access.

Some examples of TDMA are as follows;

- personal digital Cellular(PDC)
- Integrated digital enhanced network.
- Universal terrestrial radio access(UTRA)

Advantages of TDMA :

- TDMA can easily adapt to the transmission of data as well as voice communication.
- It has the ability to carry 64 kbps to 120 Mbps of data rates.
- No interference from simultaneous transmission.
- TDMA is the cost-effective technology to convert an analogue system to digital.
- Share a single carrier frequency with multiple users
- Mobile assisted handoff possible
- TDMA provides the user with extended battery life since transmitting the only portion of the time during conversations
- Flexible bit rate

CHANNELIZATION

Code-Division Multiple Access

- CDMA(code-division multiple access) is another technique used for channelization.
- CDMA technique differs from the FDMA because only one channel occupies the entire bandwidth of the link.
- The CDMA technique differs from the TDMA because all the stations can send data simultaneously as there is no timesharing.
- The CDMA technique simply means communication with different codes.
- In the CDMA technique, there is only one channel that carries all the transmission simultaneously.
- CDMA is mainly based upon the coding theory; where each station is assigned a code, Code is a sequence of numbers called chips.
- The data from the different stations can be transmitted simultaneously but using different code languages.

Advantages of CDMA

- Given below are some of the advantages of using the CDMA technique:
  - Provide high voice quality.
  - CDMA operates at low power levels.
  - The capacity of the system is higher than the TDMA and FDMA.
  - CDMA is better cost-effective.

WIRED LAN : ETHERNET STANDARDS AND FDDI

Ethernet Standards

- The Ethernet standards come under the IEEE 802 section which deal with local area networks and metropolitan area networks. In particular, IEEE 802.3 defines Ethernet.
- The different IEEE 802.3 standards define different aspects of Ethernet covering the physical layer and data link layer's media access control (MAC) of wired Ethernet.
- Some of the individual standards may introduce new versions or flavors of Ethernet to keep pace with the growing requirements for speed and performance, whereas other standards may define aspects like the data frames used.

- The different standards with their numbers are outlined in the table below:
  - **Standard Ethernet Code**
    - In order to understand standard Ethernet code, one must understand what each digit means. Following is a guide

WIRED LAN : ETHERNET STANDARDS AND FDDI

Guide to Ethernet Coding

10	at the beginning means the network operates at 10Mbps.
BASE	means the type of signaling used is baseband.
2 or 5	at the end indicates the maximum cable length in meters.
T	the end stands for twisted-pair cable.
X	at the end stands for full duplex-capable cable.
FL	at the end stands for fiber optic cable.

For example: 100BASE-TX indicates a Fast Ethernet connection (100 Mbps) that uses a twisted pair cable capable of full-duplex transmissions.

Some of Ethernet version numbering:

- 10BASE5: 10 Mb/s over coaxial cable up to 500 meters in length.
- 10BROAD36: 10 Mb/s over broadband cable, 3600 m max segments
- 10BASE-FL: 10 Mb/s fiber optic point-to-point link
- 10BASE-FB: 10 Mb/s fiber optic backbone (between repeaters). Also, known as synchronous Ethernet.

WIRED LAN : ETHERNET STANDARDS AND FDDI

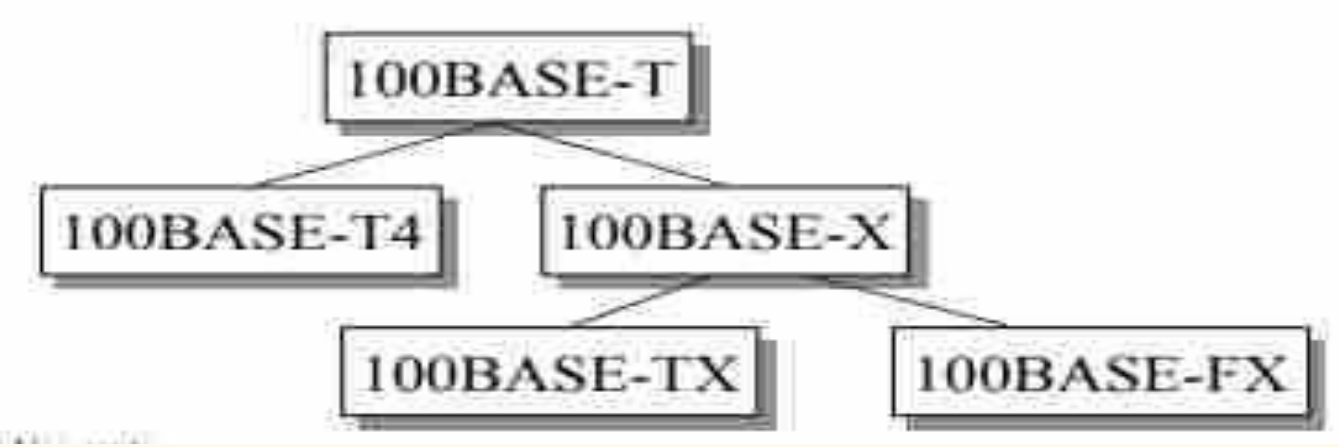
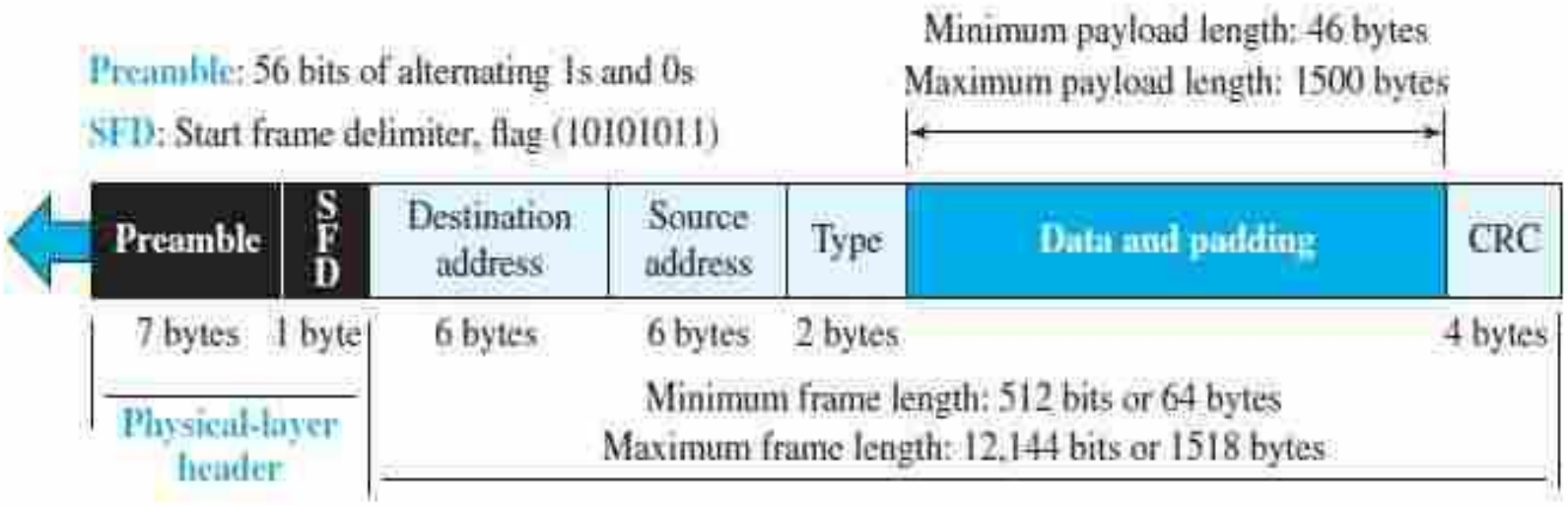


Figure 13.3 *Ethernet frame*



WIRED LAN : ETHERNET STANDARDS AND FDDI

- Basic frame format which is required for all MAC implementation is defined in IEEE 802.3 standard. Though several optional formats are being used to extend the protocol's basic capability.
- **PREAMBLE** – Ethernet frame starts (PRE).
- **Start frame delimiter (SFD).** This field (1 byte: 10101011) signals the beginning of the frame.
- **Type**. This field defines the upper-layer protocol whose packet is encapsulated in the frame.
- **Data**. This field carries data encapsulated from the upper-layer protocols. For example, a datagram has a field that defines the length(padding) of the data.
- **Cyclic Redundancy Check (CRC):** The last field contains error detection information

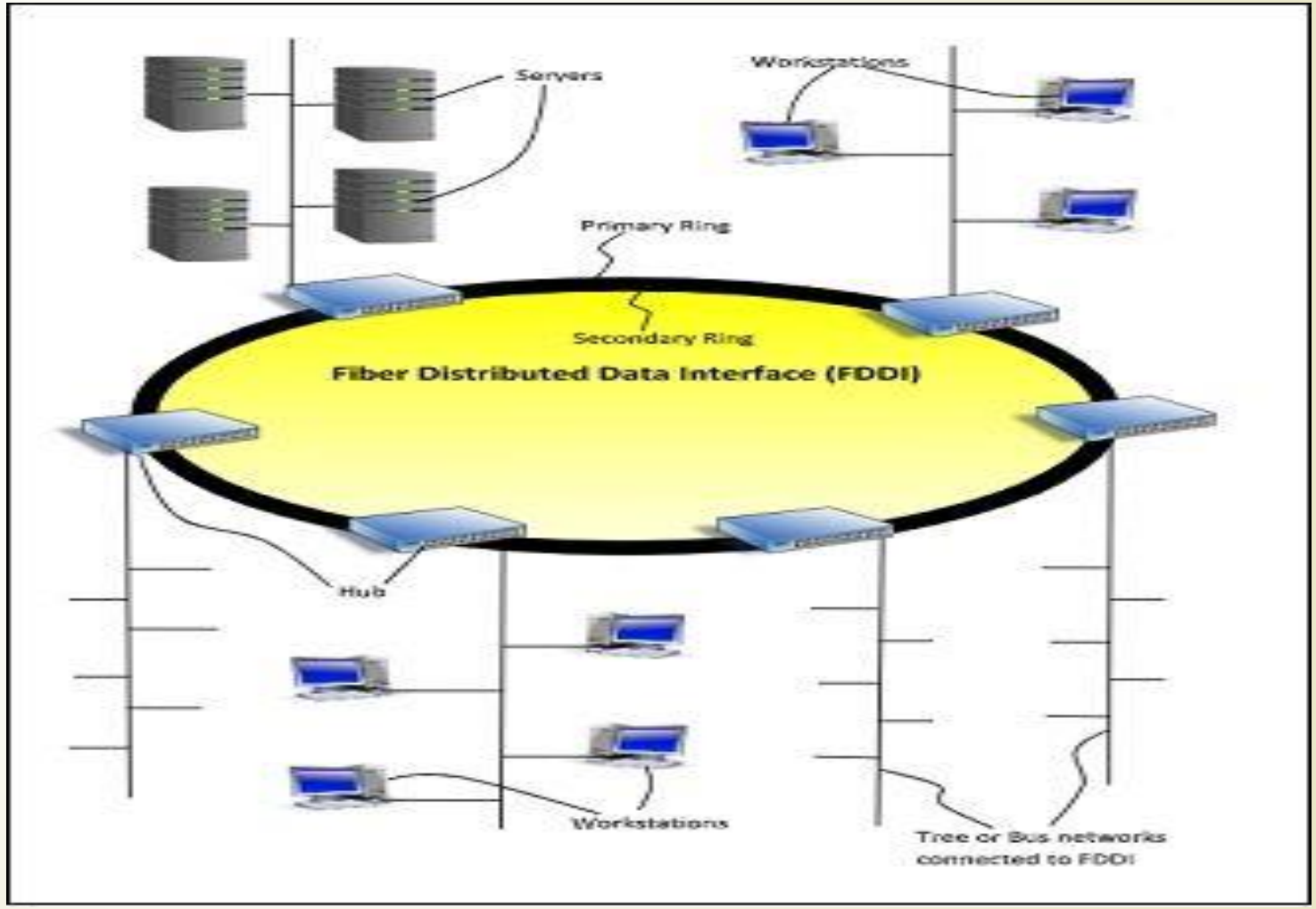
WIRED LAN : ETHERNET STANDARDS AND FDDI

Fiber Distributed Data Interface (FDDI)

- It is a set of ANSI (American National Standards Institute) and ISO (International organization for Standardization) standards for transmission of data in local area network (LAN) over fiber optic cables. It is applicable in large LANs that can extend up to 200 kilometers in diameter.
- **Features**
  - FDDI uses optical fiber as its physical medium.
  - It operates in the physical and medium access control (MAC layer) of the Open Systems Interconnection (OSI) network model.
  - It provides high data rate of 100 Mbps and can support thousands of users.
  - It is used in LANs up to 200 kilometers for long distance voice and multimedia communication.
  - It uses ring based token passing mechanism and is derived from IEEE 802.4 token bus standard.
  - It contains two token rings, a primary ring for data and token transmission and a secondary ring that provides backup if the primary ring fails.

WIRED LAN : ETHERNET STANDARDS AND FDDI

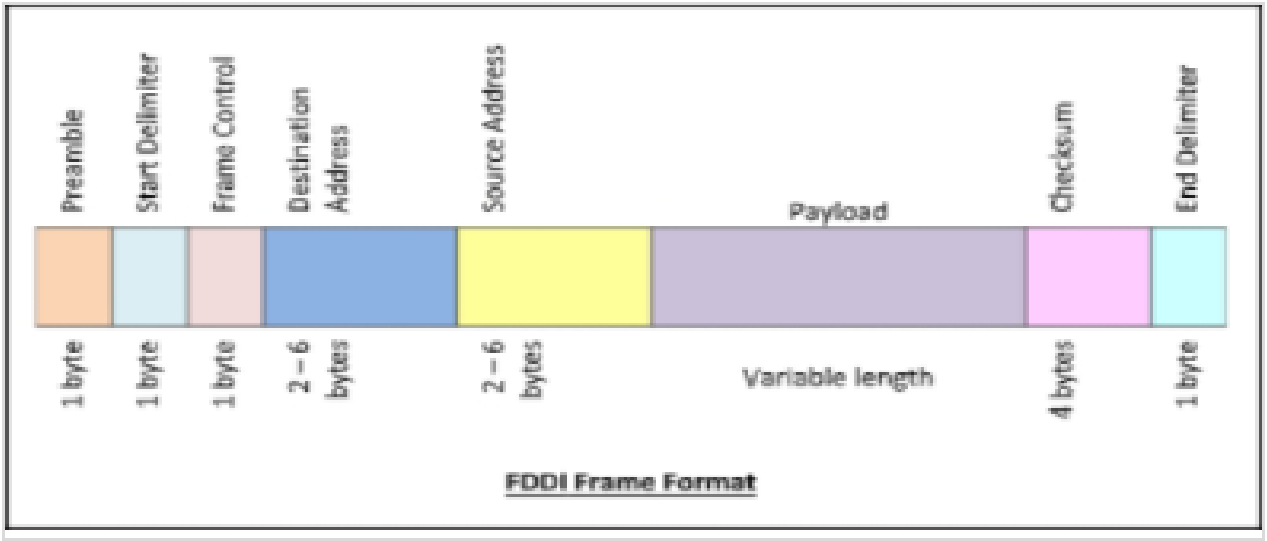
- FDDI technology can also be used as a backbone for a wide area network (WAN).
- The following diagram shows FDDI



WIRED LAN : ETHERNET STANDARDS AND FDDI

Frame Format

- The frame format of FDDI is similar to that of token bus as shown in the following diagram



WIRED LAN : ETHERNET STANDARDS AND FDDI

The fields of an FDDI frame are

- **Preamble**: 1 byte for synchronization.
- **Start Delimiter**: 1 byte that marks the beginning of the frame.
- **Frame Control**: 1 byte that specifies whether this is a data frame or control frame.
- **Destination Address**: 2-6 bytes that specifies address of destination station.
- **Source Address**: 2-6 bytes that specifies address of source station.
- **Payload**: A variable length field that carries the data from the network layer.
- **Checksum**: 4 bytes frame check sequence for error detection.
- **End Delimiter:** 1 byte that marks the end of the frame.

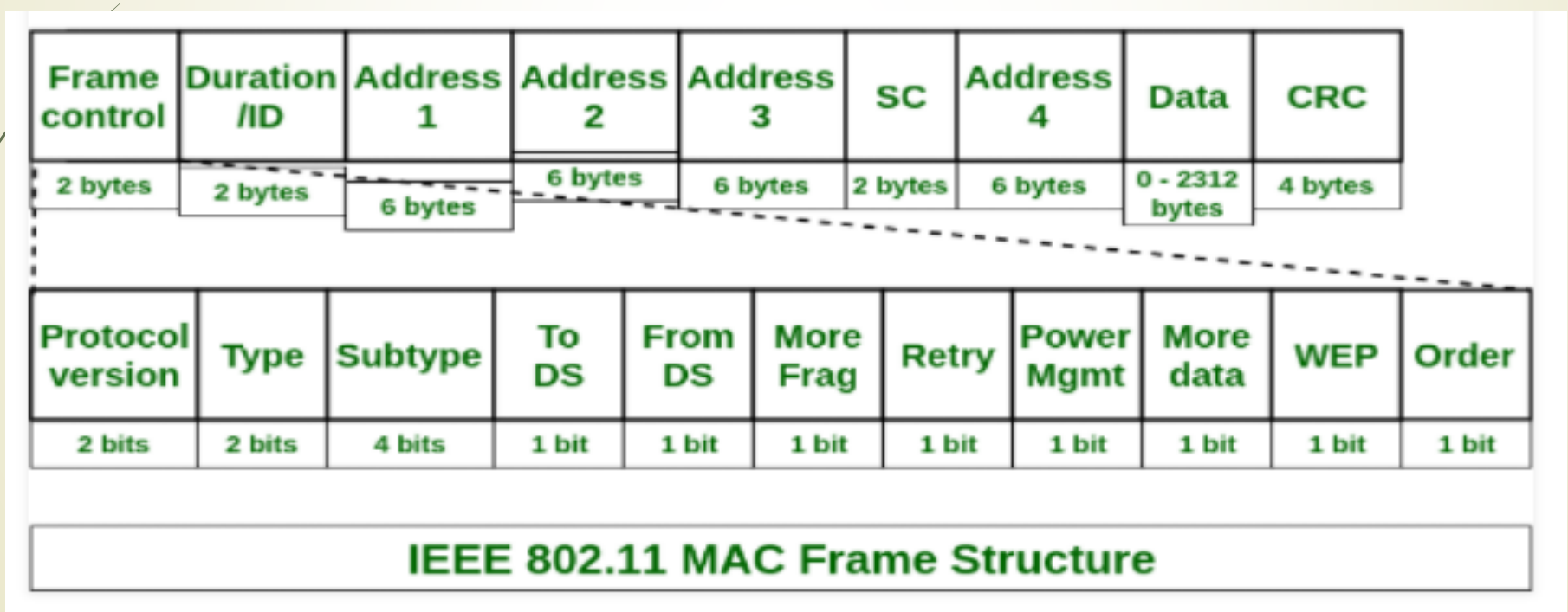
WIRELESS LAN : IEEE 802.11X

- **802.11x** is generic term to refer to the IEEE 802.11 standard for defining communication over a wireless LAN (WLAN).
- 802.11, commonly known as Wi-Fi, specifies an over-the-air interface between a wireless client and a base station or between two wireless clients
- It refers to the common flavors of Wi-Fi, most notably 802.11a, 802.11b, 802.11g, and 802.11n. IEEE 802.11 defines two MAC sub-layers :-
- **Distributed Coordination Function (DCF)** –
  - DCF uses CSMA/CA as access method as wireless LAN can’t implement CSMA/CD. It only offers asynchronous service.
- **Point Coordination Function (PCF) –**
  - PCF is implemented on top of DCF and mostly used for time-service transmission. It uses a centralized, contention-free polling access method. It offers both asynchronous and time-bounded service.

WIRELESS LAN : IEEE 802.11X

Frame Format of 802.11

- The 802.11 MAC sublayer provides an abstraction of the physical layer to the logical link control sublayer and upper layers of the OSI network.
- It is responsible for encapsulating frames and describing frame formats.
- **The MAC layer frame consists of nine fields.**



WIRELESS LAN : IEEE 802.11X

Frame Control(FC) –

- It is 2 bytes long field which defines type of frame and some control information. Various fields present in FC are:

- **Version:**
  - It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.
- **Type:**
  - It is a 2 bit long field which determines the function of frame i.e management(00), control(01) or data(10). The value 11 is reserved.
- **Subtype:**
  - It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 for beacon.
- **To DS:**
  - It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).

WIRELESS LAN : IEEE 802.11X

- **From DS:**
  - It is a 1 bit long field which when set indicates frame coming from DS.
- **More frag (More fragments):**
  - It is 1 bit long field which when set to 1 means frame is followed by other fragments.
- **Retry:**
  - It is 1 bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.
- **Power Mgmt. (Power management):**
  - It is 1 bit long field which indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.

WIRELESS LAN : IEEE 802.11X

- **More data:**
  - It is 1 bit long field which is used to indicates a receiver that a sender has more data to send than the current frame. This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.
- **WEP (Wired Equivalent Privacy)**:
  - It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.
- **Order**:
  - It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

Duration/ID

- It is 4 bytes long field which contains the value indicating the period of time in which the medium is occupied(in µs).

WIRELESS LAN : IEEE 802.11X

Address 1 to 4

- These are 6 bytes long fields which contain standard IEEE 802 MAC addresses (48 bit each). The meaning of each address depends on the DS bits in the frame control field.

SC (Sequence control)

- It is 16 bits long field which consists of 2 sub-fields, i.e., Sequence number (12 bits) and Fragment number (4 bits). Since acknowledgement mechanism frames may be duplicated hence, a sequence number is used to filter duplicate frames.

Data

- It is a variable length field which contain information specific to individual frames which is transferred transparently from a sender to the receiver(s).

CRC (Cyclic redundancy check)

- It is 4 bytes long field which contains a 32 bit CRC error detection sequence to ensure error free frame.

WIRELESS LAN : BLUE TOOTH

- IEEE 802.15
- It is a wireless LAN technology using short-range radio links, intended to replace the cable(s) connecting portable and/or fixed electronic devices.
- It is an ad hoc network where devices can automatically find each other, establish connections, and discover what they can do for each other.
- range 10-100 m.
- features are robustness, low complexity, low power and low cost.
- uses a 2.4-GHz ISM band divided into 79 channels of 1 MHz each
- A Bluetooth device has a built-in short-range radio transmitter.
- It uses Frequency Hop Spread Spectrum (FHSS) to avoid any interference.

WIRELESS LAN : BLUE TOOTH

- **Applications**
  - Automatic synchronization between mobile and stationary devices
  - Connecting mobile users to the internet using Bluetooth-enabled wire-bound connection ports
  - Dynamic creation of private networks
- Types of Bluetooth Wireless Technology
  - Depending on the power consumption and range of the device, there are 3 Bluetooth Classes as:
    - Class 1: Max Power – 100mW ; Range – 100 m
    - Class 2: Max Power – 2.5mW ; Range – 10 m
    - Class 3: Max Power – 1mW ; Range – 1 m

WIRELESS LAN : BLUE TOOTH

Protocol Architecture

- “ Bluetooth is a layered protocol architecture”
- Core protocols
- Cable replacement and telephony control protocols
- Adopted protocols

Core protocols

- Radio
- Baseband
- Link manager protocol (LMP)
- Logical link control and adaptation protocol (L2CAP)
- Service discovery protocol (SDP)

Cable Replacement protocol

- RFCOMM

Telephony control protocol

- Telephony control specification – binary (TCS BIN)

Adopted protocols

- TCP/UDP/IP
- OBEX
- WAE/WAP

WIRELESS LAN : BLUE TOOTH

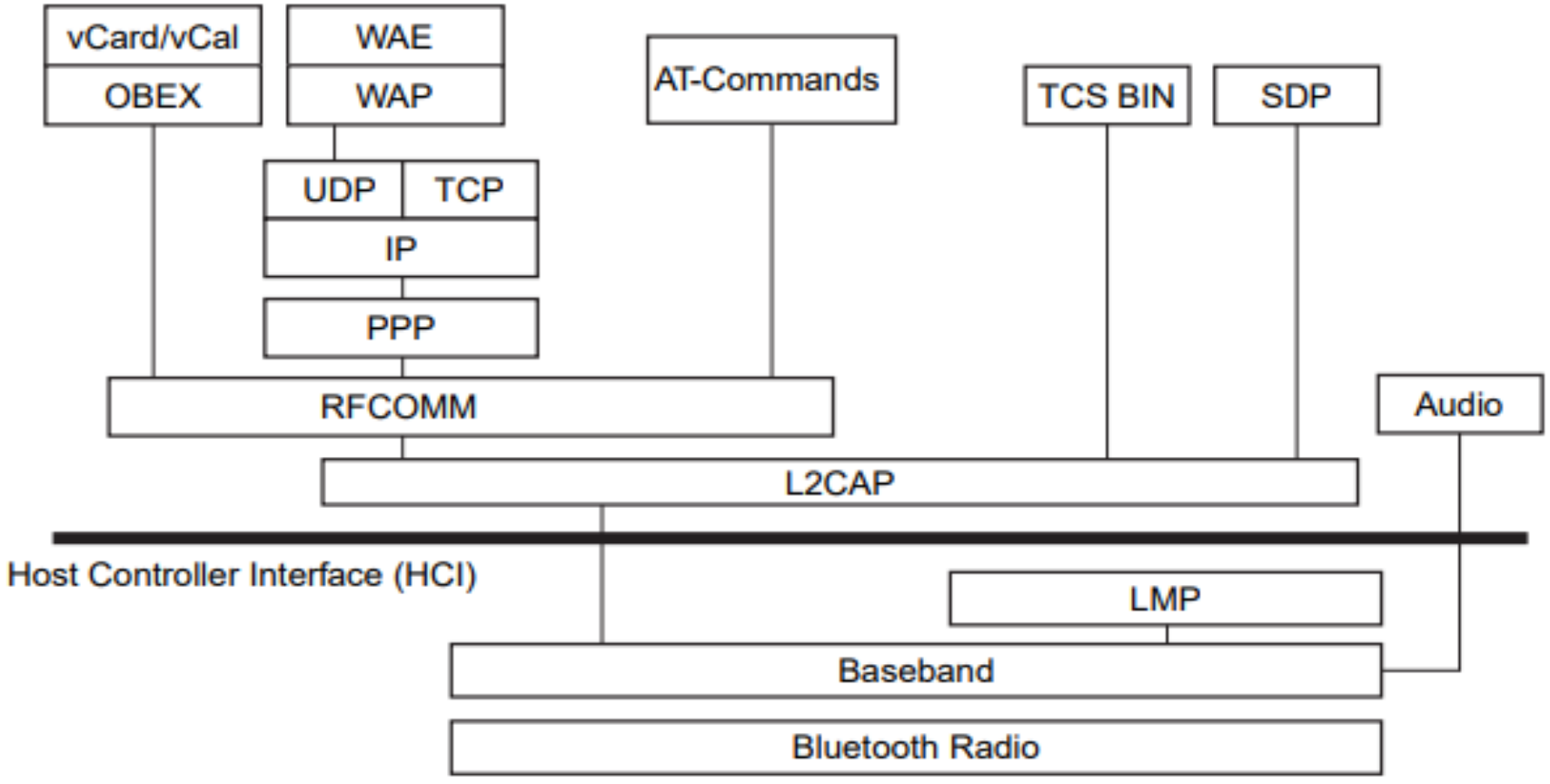


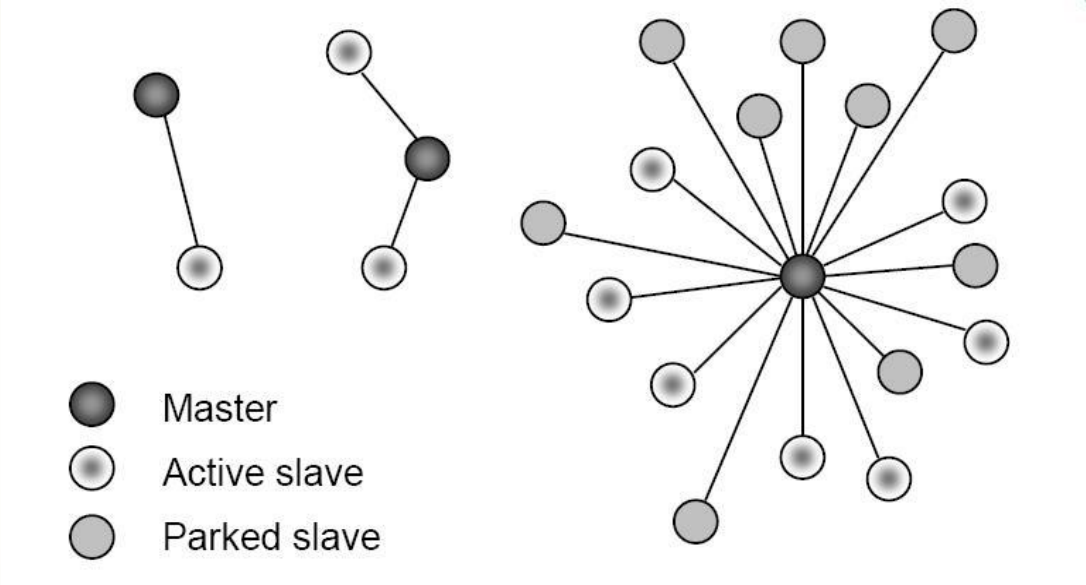
Fig.1 Bluetooth protocol stack

WIRELESS LAN : BLUE TOOTH

Bluetooth Network Topology

1. Piconet

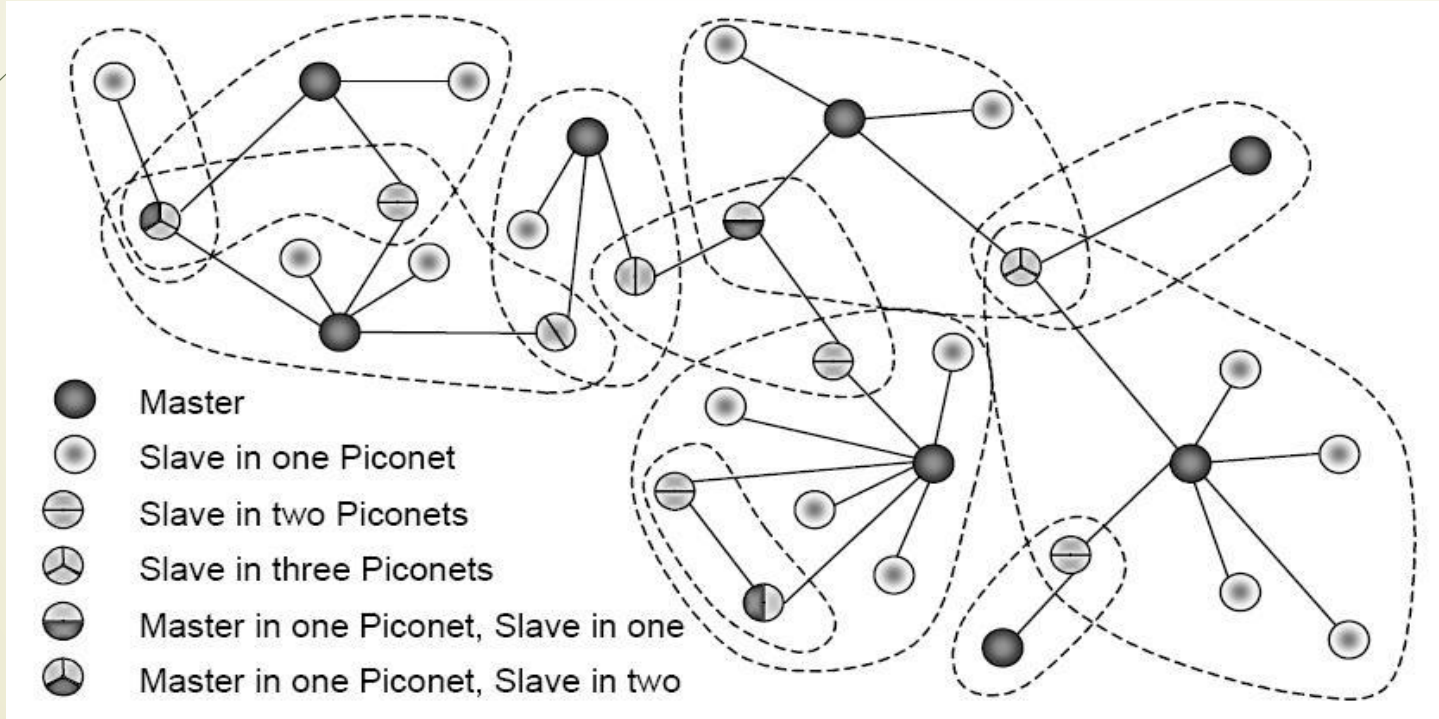
- A maximum of 8 devices (7 active slaves plus 1 master) form a Piconet
- A piconet is characterized by the master: frequency hopping scheme, access code, timing synchronization, bit rate allocated to each slave
- Only one master: dynamically selected, roles can be switched
- Up to 7 active slaves; up to 255 parked slaves
- No central network structure: “Ad-hoc” network



WIRELESS LAN : BLUE TOOTH

2. Scatternet

- Interconnected piconets, one master per piconet
- A few devices shared between piconets
- No central network structure: “Ad-hoc” network



TOKEN BUS, TOKEN RING AND VIRTUAL LAN

802.4 Token Bus

- The 802.4 IEEE standard defines the Token Bus protocol for a token-passing access method on a bus topology.
- In a token-passing access method, a special packet called a token is passed from station to station and only the token holder is permitted to transmit packets onto the LAN.
- No collisions can occur with this protocol (Only One Station can transfer)
- When a station is done transmitting its packets, it passes the token to the "next" station.
- The next station does not need to be physically closest to this one on the bus, just the next logical station.
- A station can hold the token for only a certain time.
- This assures access to all stations on the bus within a specified period of time.

TOKEN BUS, TOKEN RING AND VIRTUAL LAN

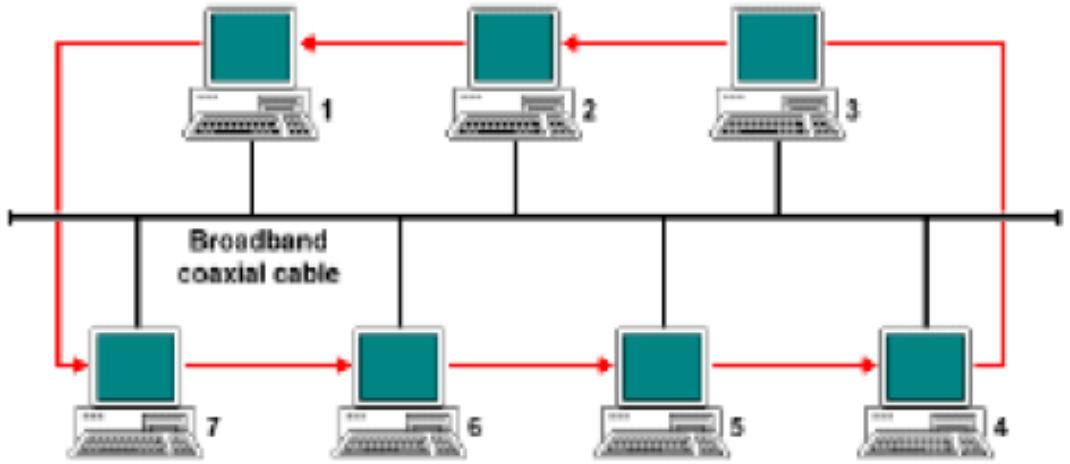


Figure : Token Bus Network (Red Arrow Indicates Token Passing Sequence)

802.5 Token

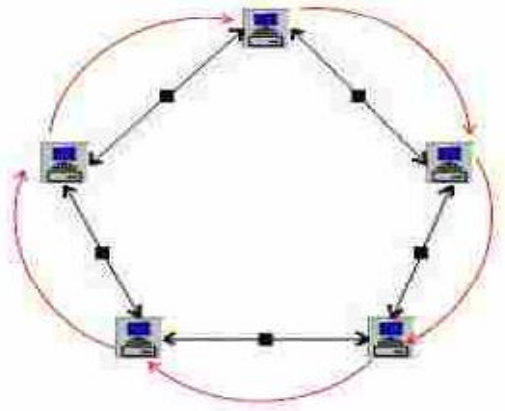


Figure : Token Bus Network (Red Arrow Indicates Token Passing Sequence)

TOKEN BUS, TOKEN RING AND VIRTUAL LAN

- The 802.5 IEEE standard defines the Token Ring protocol which, like Token Bus, is another token passing access method, but for a ring topology
- A ring topology consists of a series of individual point-to-point links that form a circle
- A token is passed from station to station in one direction around the ring, and only the station holding the token can transmit packets onto the ring
- Data packets travel in only one direction around the ring
- When a station receives a packet addressed to it, it copies the packet and puts it back on the ring
- When the originating station receives the packet, it removes the packet.

TOKEN BUS, TOKEN RING AND VIRTUAL LAN

Virtual LANs

- A VLAN is a switched network that is logically segmented by functions, project teams, or applications without regard to the physical location of users.
- For example, several end stations might be grouped as a department, such as engineering or accounting.
- When the end stations are physically located close to one another, you can group them into a LAN segment.
- If any of the end stations are in different buildings (not the same physical LAN segment), you can then group them into a VLAN.
- **Types**:
  - **Static VLAN**
    - are called port-based and port-centric membership VLANs.
    - Ports on a switch are manually assigned to a VLAN.

TOKEN BUS, TOKEN RING AND VIRTUAL LAN

- This is the most common method of assigning ports to VLANs.
- As a device enters the network, it automatically assumes the VLAN membership of the port to which it is attached

- **Dynamic VLAN**
  - allow membership based on the MAC address of the device connected to the switch port.
  - As a device enters the network, it queries a database within the switch for a VLAN membership.
  - membership is configured using a special server called a VLAN Membership Policy Server (VMPS).

- VLANs provide the following features:
  - Simplification of end-station moves, adds and changes
  - Controlled traffic activity
  - Workgroup and network security

TOKEN BUS, TOKEN RING AND VIRTUAL LAN

- **Disadvantage**:
  - Costly
  - Software based
  - Human labor to program
  - Depending on variety switches
  - Management complexity
- **Advantages**:
  - More Security
  - Ease of administration
  - Broadcast control
  - Reduction in network traffic

Your Queries Please!!!

THANK
YOU